ToySVG

Benchmarking Geometric Perception and Vector Reconstruction Under Parsimony Budgets

Benchmark Specification v0.1: Construct, Measurement, Proofs, and Validation Protocol

Mohamed Riad*
Higher School of Computer Science (ESI SBA), Sidi Bel Abbès, Algeria

6 September 2026

Abstract

Multimodal foundation models excel at describing raster images in open vocabulary, but reproducing observed visual structure as compact parametric vector geometry requires fine-grained spatial grounding under parsimony constraints. Existing vector benchmarks score rendered raster similarity or learned embeddings, conflating rasteriser antialiasing, semantic categorization, and unconstrained token length with genuine geometric perception. These proxy metrics fail to distinguish a model that perceives geometry but cannot emit coordinate tuples from one that fails perception entirely, while unconstrained command budgets reward degenerate procedural pixel transcription over structural abstraction. I introduce **ToySVG**, a deterministic benchmark that probes geometric perception and vector path reconstruction from raster pixels under strict command budgets $k \in \{16, 48, 160, \infty\}$. ToySVG evaluates blind monochrome inputs (256×256) against hidden target paths using a closed-form, arithmetic scoring pipeline combining similarity alignment, soft intersection-over-union, boundary distance, and Euler characteristic topological conservation. I prove scale-translation invariance, metric symmetry, Lipschitz continuity of the rasteriser, and false-positive bounds under randomized guessing. Across frontier vision models evaluated on the Kaggle Benchmarks platform, geometric reconstruction under budget $k = 48$ spans from $S = 0.4076$ (`claude-haiku-5-5`) to $S = 0.9419$ (`gpt-6-astra`), where top-performing reasoning architectures beat the procedural tracer ceiling of 0.7535 on 30 of 32 items. In contrast, on the perception-only control probe, models achieve up to 88.8% structure accuracy (142/160 objective fields correct for `gpt-6-astra`) and 81.2% for `gemini-3.8-flash` (100% on bilateral symmetry), empirically demonstrating that spatial perception precedes and decouples from parametric coordinate synthesis under parsimony budgets. All evaluation metrics, seeds, and validity thresholds are pre-registered, and the closed-form scoring pipeline executes deterministically across independent computing environments.

Contents

1 Introduction **3**
1.1 Task Formulation 3
1.2 Figure Reconstruction as a Perceptual Probe 4
1.3 Construct decomposition 4
1.4 Design commitments 5
1.5 Document contributions 5

*Email: `m.riad@esi-sba.dz`

2 Position within prior work **5**
2.1 Raster-to-vector algorithms and foundation models . . . 5
2.2 Vector graphics benchmarks . . . 6
2.3 Instruction editing and semantic judges . . . 6
2.4 Operational Positioning and Comparative Distinctions . . . 6
2.5 Multimodal Perceptual Bounds . . . 7
2.6 Psychometric Calibration and Construct Validity . . . 7
2.7 Neuropsychological Visuomotor Protocols . . . 7
2.8 Comparative Gap Analysis . . . 7

3 Task specification **7**
3.1 Notation . . . 7
3.2 Functional Task Formulation . . . 7
3.3 Output Contract and Execution Rules . . . 8
3.4 Command Budget Tiers . . . 9
3.5 Validity Gating and Outcome Classification . . . 9
3.6 Prompt Protocol . . . 9

4 Ground-truth corpus **10**
4.1 Asset Source, Licensing, and Repository Pinning . . . 10
4.2 Geometric Regularity and Analytic Ground-Truth Properties . . . 10
4.3 Structural Representations across Pipeline Stages . . . 11
4.4 Complexity Stratification . . . 13
4.5 Three-Way Corpus Partitioning and Identification Logic . . . 13
4.6 Sampling Protocol and Stratum Balance . . . 13
4.7 Corpus Hygiene and Security Discipline . . . 13

5 Measurement pipeline and metrics **14**
5.1 Pipeline Architecture and Operational Flow . . . 14
5.2 Deterministic Rasterisation Pipeline . . . 14
5.3 Spatial Alignment and Similarity Transformations . . . 14
5.4 Component Metric Formulations . . . 15
5.5 Geometric Behavior on Synthetic Reference Pairs . . . 15
5.6 Reference Anchors and Normalization Baselines . . . 18
5.7 Statistical Estimation and Resampling Protocol . . . 19
5.8 Empirical Worked Example on Benchmark Items . . . 19
5.9 Detailed Numerical Walkthrough . . . 19
5.10 Score Computation: Explicit Evaluation Trace . . . 20
5.11 Metric Behavior on Empirical Model Generations . . . 22
5.12 Empirical Kaggle Evaluation: Multi-Model Benchmark . . . 23
5.13 Qualitative Visual Plate Analysis . . . 26

6 Properties and proofs **26**
6.1 Lipschitz Continuity of Signed Distance Fields . . . 26
6.2 Rasterisation Discretization Error Bounds . . . 29
6.3 Identity Exactness and Metric Reflexivity . . . 29
6.4 Score Bounds and Component Inferences . . . 29
6.5 Boundary Metrics via Distance Field Envelopes . . . 30
6.6 Topological Betti Number Invariants . . . 30
6.7 Well-Posedness and Monotonicity of Budget Ceilings . . . 31
6.8 Similarity Alignment Convergence and Residual Bounds . . . 31
6.9 Identification of Parametric Recall . . . 31
6.10 Statistical Power and Sample Size Requirements . . . 32

6.11 Item Response Theory Calibration 32
6.12 Automated Invariant Verification Battery 33

7 Threats to validity, leakage, and gaming 33
7.1 Threat Modeling and Countermeasure Taxonomy 33
7.2 Delimited Construct Scope and Non-Goals 33

8 Validation plan 34
8.1 Construct Validity via Formal Invariant Batteries 34
8.2 Delimited Validity Scope and Subjective Aesthetics 35
8.3 Replication Reliability and Execution Invariance 35
8.4 Sensitivity and Monotonicity Envelopes 36
8.5 Parsimony Profiles across Budget Tiers 36
8.6 Blind Structural Control Diagnostic 36
8.7 Metric Ablation and Rank Correlation Invariance 37
8.8 Degradation and Resolution Robustness 37

9 Phases and exit gates 38
9.1 Execution Phase Specifications 38
9.2 Saturation Monitoring and Procedural Refresh 38

10 Reporting standard 39
10.1 Canonical Benchmark Leaderboard 39

11 Limitations 40

12 Open questions 41

13 Harness Reproduction Protocol 41

A Scanline Winding Algorithms and Topological Correctness 41

B Prompt Templates and Instruction Contracts 42

C Glossary 43

1 Introduction

1.1 Task Formulation

A model is shown one monochrome raster image. The image is the rendering of a path d that the model never observes directly. The model must emit path data, in the same coordinate frame, that reproduces the figure. The benchmark scores the emitted geometry against the hidden original with a deterministic, fully reproducible pipeline.

Existing vector benchmarks measure image similarity over rendered bitmaps, conflating renderer antialiasing with geometric accuracy while leaving path complexity unconstrained. By contrast, ToySVG evaluates the emitted path directly against the hidden ground truth under a formal drawing command budget.

Three design choices distinguish this formulation from existing vector-generation evaluations:

1. **Blind input.** The prompt contains no icon name, category, file name, or caption. The file identifier on disk is a salted SHA-256 hash, removing textual cues that could trigger parametric recall of specific training assets.

2. **Command budget.** The predicted path must use at most k drawing commands, where $k \in \{16, 48, 160, \infty\}$. A naive pixel tracer reproduces raster contours faithfully with thousands of commands; the budget ceiling prevents this transcription strategy. Every evaluation tier is scored against the best deterministic tracer fitting the identical budget, isolating how much achievable fidelity the model captures relative to procedural tracing.
3. **Deterministic geometric scoring.** Headline evaluation derives from region overlap, boundary distance, and topological invariants. Perceptual and embedding metrics appear exclusively in ablation studies, because geometrically divergent drawings can register adjacent in embedding space, while accurate renderings are penalized by discretization and antialiasing artifacts.

1.2 Figure Reconstruction as a Perceptual Probe

Scoring one item costs one rasterisation and three comparisons on a 256×256 canvas. That pipeline runs on one CPU core, so evaluation cost does not grow with the parameter count of the system under test. The released pack holds 32 items, and a full evaluation runs on a CPU in one command.

The score is a pure function of the two paths. A parse, a rasterisation with a pinned renderer, one similarity fit, and three comparisons return the same number for the same answer on every machine. A published figure can therefore be re-derived from the released pack without re-running the model that produced it, which is what makes a reported number checkable rather than trusted.

That construction also makes copying a dense perceptual probe. Vision-language models exhibit documented weaknesses on elementary spatial operations: counting overlapping shapes, tracking continuous paths, and detecting geometric intersections [14]. Most visual question-answering benchmarks query discrete categories that language priors can frequently infer. A vector path carries more information than a category label: a path specified by 48 commands defines dozens of constrained coordinate tuples that cannot be hallucinated from the textual prompt, and the blind input removes the icon name that would make recall possible.

The task builds on classical methodology from neuropsychology. The Beery Visual-Motor Integration assessment [2] and the Rey–Osterrieth Complex Figure test [23] evaluate visuomotor execution by requiring subjects to copy geometric figures, scoring accuracy and structural placement independently. This literature treats figure copying as a measure of perceptual-motor execution, and it supplies an established measurement protocol: decompose figures into structural units, score local geometry and spatial placement separately, and report the full decomposition. I adopt this protocol, replacing clinical scoring with an automated deterministic pipeline.

1.3 Construct decomposition

Evaluating whether a model can perceive and draw conflates distinct cognitive operations. I measure these capabilities along four orthogonal axes:

Perception. Does the model encode the spatial geometry present in the input raster? Operationally: can the model answer structural queries regarding connected components, hole counts, and orientation axes?

Abstraction. Can the model extract a minimal set of parameterized primitives capturing the figure, rather than transcribing pixel contours? Operationally: the performance gradient across budget tiers k, and the parsimony ratio relative to the tracer ceiling at k.

Reproduction. Can the model emit syntactically valid vector geometry matching the hidden original? Operationally: the geometric composite metric evaluated in the unconstrained $k = \infty$ tier.

Parsimony and control. Does the generation respect operational formatting constraints? Operationally: a set of deterministic validity gates enforcing XML parsing, budget compliance, and bounding-box canvas containment.

The benchmark evaluates monochrome vector geometry. Colour, gradients, text, typography, artistic styling, and semantic categorization belong to distinct evaluation instruments; folding these disparate attributes into a single scalar obscures diagnostic failure modes.

1.4 Design commitments

C1. Bit-for-bit determinism. The score is a pure mathematical function of ground-truth path and emitted output string. The rasteriser is committed directly within the repository rather than imported as an external dependency. Identical inputs yield identical byte outputs across computing environments.

C2. Absence of subjective judges. No language models or human raters are employed in scoring, validation, or ranking. Because the target is a hidden ground-truth geometry, fidelity is computed directly, and validation reduces to invariance, synthetic order recovery, and convergence tests (§8.1).

C3. Exposed decomposition. The benchmark reports every constituent metric alongside composite scores, distinguishing whether performance stems from global shape overlap, local boundary alignment, or topological recovery.

C4. Bounded empirical anchors. Baseline floors (empty output, bounding boxes), algorithmic ceilings (procedural tracing), and perturbation anchors (translated copies) are evaluated alongside model outputs, bounding the operational metric interval.

C5. Principled leakage isolation. The corpus splits items to identify training-set memorization quantitatively, reporting memorization gaps alongside benchmark accuracy.

C6. Pre-registration discipline. Metrics, weighting coefficients, splits, and evaluation protocols are formalized prior to large-scale data collection [12].

C7. Provenance tracking. Every test item maps to a pinned release of an open-source icon set with verified licensing, and all datasets are indexed by cryptographic content hashes.

1.5 Document contributions

- A task formalization separating visual perception from vector code generation (§3).
- A corpus split design providing statistical identification of parametric recall (§4, §6.9).
- A pinned, deterministic measurement pipeline with formal error bounds and invariant proofs (§5, §6).
- A validation protocol with power derivations and item-response calibration (§8).
- An execution roadmap governed by empirical numerical exit criteria (§9).

2 Position within prior work

2.1 Raster-to-vector algorithms and foundation models

Converting bitmaps to parametric vector paths has historically relied on algorithmic contour tracing (such as Potrace [19]) or specialized optimization-based inverse graphics pipelines. While algorithmic tracers reproduce raster contours with high visual fidelity when command budgets are unconstrained, they operate mechanically without spatial abstraction or semantic parsimony.

ToySVG addresses a fundamentally different evaluation target: assessing whether general-purpose multimodal foundation models possess genuine spatial perception and the capacity to synthesize compact parametric vector geometry under strict command budgets. Classical deterministic tracing is retained strictly as an algorithmic upper bound (T_k), defining the achievable geometric fidelity for a given raster at budget k.

2.2 Vector graphics benchmarks

VGBench [24] evaluates language models on SVG, TikZ, and Graphviz code generation conditioned on text captions, scoring outputs with CLIP similarity and Fréchet Inception Distance (FID). Both metrics operate on corpus-level distributions: they reward general semantic proximity while failing to register local geometric deviations, and FID cannot score individual test instances. Understanding tasks in VGBench rely on multiple-choice formats susceptible to language priors.

SVG-Bench, introduced alongside StarVector [17], evaluates image-to-SVG generation on individual items across ten pooled datasets. The evaluation relies on raster-to-raster comparison: both ground-truth and predicted SVGs are rendered through CairoSVG on a fixed canvas, evaluating Mean Squared Error (MSE), SSIM, LPIPS, and DinoScore between resulting bitmaps [17, 16]. Ground-truth geometry enters the evaluation solely as a rasterized projection rather than parametric path data, omitting explicit spatial alignment. Non-compiling predictions are mapped to blank white canvases rather than diagnostic validity gates.

Three properties of SVG-Bench inform my design. First, while [17] demonstrates that MSE and SSIM misrank vector graphics, replacing them with DINOv2 embeddings forfeits metric auditability and determinism. DinoScore relies on a specific opaque checkpoint, exhibiting documented discrepancies between published descriptions (L_2 distance) and released code (cosine similarity rescaled to $[0, 1]$) [16]. Second, top leaderboard entries reflect specialist checkpoints fine-tuned on benchmark splits and evaluated via best-of-five selection on the reported metric, measuring inference-time search rather than single-pass generation. Third, reported DinoScore values compress into $[0.94, 0.98]$ across model families [17], providing insufficient dynamic range to resolve structural distinctions.

From SVG-Bench I adopt image-conditioned framing, item-level granularity, and explicit disclosure of parsing failures. ToySVG diverges by evaluating parametric path data directly against artist paths, fitting explicit similarity transforms, enforcing hard budget tiers, publishing deterministic tracer ceilings, and eliminating learned weights from the scoring metric.

2.3 Instruction editing and semantic judges

A second research thread examines instruction-based vector editing. SVGEditBench [11] evaluates six synthetic edit templates (color, contour, cropping, rotation) over emoji assets using raster metrics. SVGenius [3] broadens the scope to multimodal editing under algorithmic mutations. VectorEdits [9] constructs 271,306 edit pairs by mining visually similar vector pairs within icon collections and captioning differences using vision-language models.

The VectorEdits construction exhibits an instructive structural limitation. Because pairs are matched by visual proximity rather than produced by verified transformations, the reported delta is inferred rather than ground truth. Furthermore, relying on unnormalized CLIP and DINOv2 similarities yields a degenerate failure mode: a null baseline returning the input raster unchanged scores 0.9634 CLIP and 0.9011 DINOv2, exceeding every trained editor evaluated [9]. On that metric, null identity generation wins the leaderboard, requiring baseline floor subtraction before meaningful comparison is possible.

VectorGym [18] evaluates four editing tasks with expert annotations, combining raster metrics with a multimodal model judge. Independent analysis indicates that this judge's agreement with human raters reaches Spearman $\rho = 65.1$ [4], trailing standard zero-shot multimodal models (67.8, 67.1, 67.0). SVG-Score [4] further demonstrates that CLIP metrics remain insensitive to component counts and spatial relations on monochrome graphics. When evaluating black-and-white vector graphics, embedding metrics frequently fail to register severe topological contradictions.

2.4 Operational Positioning and Comparative Distinctions

Prior evaluations compare rendered raster outputs: pixel differences in SVG-Bench, embedding similarities in VectorEdits, and language-model judgments in VectorGym. None evaluates predicted

commands directly against the underlying parametric ground-truth path. Consequently, existing benchmarks cannot differentiate a boundary translated by three pixels from a filled component of equivalent area. Furthermore, learned evaluators require heavy neural checkpoints, leaving substantial variance unexplained while introducing non-deterministic model dependencies.

ToySVG evaluates monochrome Phosphor icon paths directly against original ground-truth vector geometry. The evaluation pipeline combines explicit floor baselines, per-item tracer ceilings, deterministic operational validity gates, and closed-form geometric metrics free of learned parameters.

Two limitations warrant explicit statement. First, scale: SVG-Bench incorporates test splits spanning 96 to 5,700 items; the current ToySVG evaluation suite comprises a compact, curated corpus. Second, recall control: identifying parametric memorization relies on three-way corpus partitioning between clean in-distribution icons, degraded variants, and procedural out-of-distribution paths.

2.5 Multimodal Perceptual Bounds

Multimodal foundation models exhibit documented blind spots on basic spatial reasoning: counting overlapping contours, determining path intersections, and parsing rotated characters [14]. This empirical limitation indicates that figure copying failures frequently originate in visual perception rather than vector syntax generation. This finding motivates the structural control task in §8.6: probing whether a model can perceive regions, holes, and symmetry axes independently of code synthesis.

2.6 Psychometric Calibration and Construct Validity

Empirical benchmark rankings are sensitive to task composition and aggregation schemes [5]. I mitigate this instability by freezing metric weights in advance and reporting stratified performance. Construct validity requires formal verification rather than informal assertion [7, 1]. I formalize item difficulty and discrimination parameters using item-response theory [15, 13], reporting specifications in accordance with standardized documentation frameworks [6, 10].

2.7 Neuropsychological Visuomotor Protocols

Clinical assessments such as the Beery VMI [2] and Rey–Osterrieth Complex Figure [23] score copying accuracy and structural placement as orthogonal dimensions. I transfer two methodological principles directly to ToySVG: first, isolating spatial alignment before evaluating local boundary curvature (§5.3); second, penalizing topological errors (merged components, missing holes) through dedicated structural metrics rather than undifferentiated area overlap.

2.8 Comparative Gap Analysis

3 Task specification

3.1 Notation

3.2 Functional Task Formulation

Definition 1 (Figure-copying task)**.** A task instance is a tuple (ι, d): a monochrome image $\iota : \Omega \to [0, 255]$ representing the rasterization of hidden path d, where d defines closed filled region $R \subset \Omega$. A model is a deterministic or stochastic map $M : \mathcal{I} \to \mathcal{S}$ from raster images $\mathcal{I}$ to string sequences $\mathcal{S}$. The evaluation evaluates pushforward metric

$$S\,(R,\ M(\iota))\ \in\ [0, 1],$$

where S denotes the deterministic functional formalized in §5. The model input excludes path data d, semantic labels, icon names, and textual metadata.

Table 1: Positioning against nearest prior instruments. Recall denotes control for parametric memorization; budget denotes explicit path complexity limits; ceiling denotes deterministic upper bounds on achievable fidelity; reference format distinguishes parametric path evaluation from raster comparisons; learned terms mark dependencies on neural checkpoints.

	VGBench	**SVG-Bench**	**Traced SVG**	**ToySVG**
Input modality	caption	image	image	image
Items	5,845	10 datasets	arbitrary	Curated pack
Unit of scoring	corpus	per item	per item	per item
Reference format	None	Raster bitmap	Raster bitmap	Parametric path
Headline metric	CLIP + FID	DinoScore	Pixel error	Geometric composite
Learned score terms	Yes	Yes	No	No
Spatial alignment	Viewport only	Viewport only	None	Similarity fit
Complexity budget	None	Reported only	None	Four discrete tiers
Tracer ceiling published	No	No	None	Yes
Recall identification	None	Splits only	None	Three-way partition
Validity gating	None	Post-processed	None	Hard operational gates
Metric validation	None	Human preference	None	Formal protocol

Table 2: Mathematical notation. Pixel coordinates index the discrete analysis canvas of dimension $n \times n$ with $n = 256$.

Symbol	Mathematical Interpretation
Ω	Discrete analysis canvas, $\Omega = [0, n)^2 \subset \mathbb{R}^2$ with $n = 256$
$R \subset \mathbb{R}^2$	Ground-truth filled region corresponding to hidden path d
$P \subset \mathbb{R}^2$	Prediction region parsed from output and aligned to R
$A, B : \Omega \to [0, 1]$	Coverage functions (fractional pixel area) of R and P
$\varphi(z)$	Signed distance function to boundary ∂R, with $\varphi < 0$ on $\text{int}(R)$
∂R	Continuous topological boundary of ground-truth region
k	Drawing command budget constraint
$S \in [0, 1]$	Deterministic composite geometric score
$s_\bullet$	Constituent metrics: s_R (region), s_E (boundary), s_T (topology)
T_k	Deterministic procedural tracer constrained to budget k
$\mathcal{F}_k$	Set of valid SVG path strings containing at most k drawing commands
$C(k)$	Algorithmic tracer ceiling, $C(k) = S(R, \mathsf{T}_k(\iota))$
σ	Flattened coverage vector of dimension n^2

Remark 2 (Determinism as Formal Metric Invariance). Metric validity requires that evaluation functionals remain invariant to software runtime environments. If scores vary across renderer implementations or library versions, leaderboard rankings reflect environment artifacts rather than model capabilities. Fixing S as a self-contained, pinned mathematical function guarantees bit-for-bit reproducibility across independent computing platforms.

3.3 Output Contract and Execution Rules

Models emit text strings. The parsing module accepts either a raw SVG path string d or a valid XML document containing one or more `<path>` elements. The execution contract fixes the following semantics:

- **Coordinate system.** Path coordinates lacking explicit `viewBox` declarations evaluate in canvas space $[0, 256]^2$. Documents declaring an explicit `viewBox` map to canvas bounds via uniform aspect-ratio scaling and centered translation.
- **Rasterization semantics.** Filled interiors and stroked trajectories contribute ink to canvas coverage. Trajectories declaring explicit stroke widths undergo Minkowski dilation with circular

kernels of corresponding radii prior to rasterization.

- **Winding rules.** Path geometries default to the `nonzero` winding rule unless explicitly overridden by an `evenodd` attribute.
- **Paint interpretation modes.** Under W3C SVG 1.1 specifications, path elements declaring stroke attributes without explicit fill definitions are rendered with filled interiors [21]. To evaluate both syntactic strictness and intended geometry, I compute two parallel metrics: *spec-compliant* (literal W3C fill defaults) and *tolerant* (treating stroke-only elements as hollow contours). Both metrics are reported in full.
- **Single-attempt evaluation.** Evaluation proceeds without verifier loops, multi-turn self-correction, or execution feedback. Generation defaults to greedy decoding ($T = 0$) on single attempts.

3.4 Command Budget Tiers

Budgets are measured by counting drawing operators emitted in the candidate path: each instance of `M`, `L`, `C`, `S`, `Q`, `T`, `A`, and `Z` consumes one command unit. Coordinate arguments do not increment the count; cubic Bézier segments incur the identical budget cost as linear segments, incentivizing compact parametric representation over dense polyline approximations.

Table 3: Command budget allocations. Intermediate tiers B1 and B2 isolate structural abstraction; tier B3 provides an unconstrained baseline.

Tier	Budget k	Designation	Diagnostic Target
B0	16	Sparse	Coarse silhouette abstraction and primary orientation
B1	48	Structural	Component decomposition, genus, and relative proportions
B2	160	Detailed	Local boundary curvature, junctions, and high-frequency features
B3	∞	Unbounded	Asymptotic fidelity ceiling under unconstrained tracing

3.5 Validity Gating and Outcome Classification

Diagnostic fidelity requires classifying failure modes explicitly. Every evaluated item receives a deterministic outcome code. Failed items receive a headline score of zero while remaining in aggregate denominators to prevent optimistic selection bias.

Table 4: Diagnostic outcome taxonomy. Partial evaluations record geometric metrics separately for failed generations to audit error causes.

Code	Headline S	Operational Criterion
`ok`	S	Syntax valid, budget respected, canvas bounds satisfied
`empty`	0	Empty output or whitespace string without path syntax
`unparsable`	0	Path syntax rejected by strict validating parser
`degenerate`	0	Valid syntax yielding measure zero on Ω (no positive area)
`over_budget`	0	Emits command count exceeding budget k (partial score logged)
`off_canvas`	0	All emitted geometry lies outside canvas boundary Ω
`no_svg`	0	Natural language refusal or non-SVG formatting
`truncated`	0	Generation terminated by maximum output token limits

3.6 Prompt Protocol

Full prompt specifications are listed in Appendix B. The prompt adheres to four operational principles:

1. **Syntax isolation.** The prompt provides a minimal syntax specification using an elementary geometric primitive absent from the corpus, preventing XML parsing errors from masquerading as perceptual deficits.
2. **Metadata exclusion.** Prompts omit asset identifiers, semantic category labels, and descriptive captions, eliminating textual recall cues.
3. **Verbatim template uniformity.** All models receive identical instruction strings; temperature variations and prompt ablations are evaluated as distinct experimental configurations.
4. **Explicit constraint specification.** Prompts declare the exact integer command limit k and define operator counting rules directly.

4 Ground-truth corpus

4.1 Asset Source, Licensing, and Repository Pinning

The corpus derives from **Phosphor Icons**, core release version 2.1.1, under the permissive **MIT** license. The release contains 1,512 icons across six weight variants (`thin`, `light`, `regular`, `bold`, `fill`, `duotone`), requiring 6.5 MB uncompressed. The MIT license grants unrestricted commercial and academic usage conditioned on attribution retention; license terms are redistributed alongside corpus manifests. Table 5 summarizes alternative candidate libraries evaluated during selection. Selection criteria required permissive licensing, strict geometric uniformity (§4.2), and archival footprints suitable for deterministic content-hashing.

Table 5: Icon asset libraries evaluated for corpus integration. Licensing terms reflect package metadata published on npm registries as of September 2026.

Project	Licensing Specification and Geometric Characteristics
Phosphor Icons 2.1.1	**MIT (Selected)**. Uniform 256×256 coordinate canvas; single closed filled path for five of six weights
Tabler Icons 3.47	MIT; stroke-based primitives requiring ambiguous stroke-to-fill runtime expansion
Lucide 1.47	ISC; stroke-based primitives on non-integer 24-unit viewports
Material Symbols 0.47	Apache-2.0; variable-font derivation containing multi-scale layout redundancies
Simple Icons 16.31	CC0-1.0; proprietary corporate trademarks posing jurisdictional fair-use restrictions
Font Awesome 7.3	Compound CC-BY-4.0 / OFL-1.1 / MIT attribution split across subdirectories
Iconify JSON 2.2	MIT aggregator (484 MB uncompressed); candidate for multi-family generalization splits

4.2 Geometric Regularity and Analytic Ground-Truth Properties

Every selected asset adheres to the canonical XML structure:

```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 256 256" fill="currentColor">
  <path d="..."/>
</svg>
```

with `currentColor` mapping to monochrome black ink. Five weight strata define geometry using exactly one `<path>` element; the `duotone` variant uses two elements with secondary opacity set to 0.2. This structural homogeneity provides an explicit mathematical advantage: the reference region corresponds to an exact parametric path on integer grid $[0, 256]^2$. Ground truth is re-rasterized at runtime through the pinned renderer (§5.2), eliminating rasterization discrepancies and anti-aliasing artifacts from external toolchains.

4.3 Structural Representations across Pipeline Stages

Each corpus item transitions through three distinct operational representations, isolating evaluation from external side effects.

Table 6: Operational representations per corpus instance. Evaluating parsed vector paths rather than bitmap diffs guarantees coordinate-frame consistency.

Object	Concrete Format	Functional Role in Evaluation Pipeline
Model Input	256×256 8-bit grayscale PNG	Rendered from vector geometry at $4\times$ supersampling without metadata or coordinate cues; provides visual stimuli.
Hidden Ground Truth	Parametric path string `d`	Recorded in `data/gt/<item_id>.json`; re-rasterized at scoring time via pinned rasterization routines (§5.2).
Predicted Answer	Output string: bare `d` or XML	Parsed into discrete geometric primitives via the identical parser applied to ground truth.

Metadata and Ground-Truth Record Structure. Each test item maps to a self-contained JSON schema:

$$\underbrace{\texttt{gt_d}}_{\text{target path}}\texttt{, gt_fill_rule, n_commands_gt, n_subpaths_gt,}$$

$$\texttt{topology, ink_area_px, stroke_median_px,}$$

accompanied by asset provenance tags (`name`, `weight`, `source_sha256`) and benchmark partition metadata (`item_id`, `split`, `stratum`, `budget`, `canvas`, `seed`). While scoring requires only `gt_d` and `gt_fill_rule`, secondary fields allow independent re-computation of complexity invariants.

Ground-Truth Representation and Verbatim Preservation. Benchmark evaluation scores predictions directly against the author's published path string. For each item, stored field `gt_d` preserves the exact character sequence of the source asset, rather than an approximated or re-simplified polygon. Table 7 details characteristics across the demonstration set, recording string lengths, command counts, and subpath totals.

Table 7: Demonstration pack ground-truth parameters. Command budgets reflect parser units counted on raw path expressions.

Item	Asset Identifier	Partition Split	Stratum	Characters	Commands
01001100	currency-rub	ID-CLEAN	S	169	25
014164b6	user-switch	ID-CLEAN	M	609	41
099fb352	plugs-connected	ID-CLEAN	C	848	66
0a3854c1	piggy-bank	ID-CLEAN	C	739	58
1009c27d	axe	ID-CLEAN	M	668	36
108207a5	circles-three	ID-CLEAN	S	286	24
15882466	git-fork	ID-CLEAN	M	311	31
16f8e20c	slack-logo	ID-CLEAN	C	535	58
1decaf83	sidebar-simple	ID-CLEAN	S	135	21
22b84da3	box-arrow-down	ID-CLEAN	M	375	38
20a8ff8d	arrow-left	ID-DEGRADED	S	148	13
14aa9d21	eyedropper-sample	OOD-DISJOINT	C	728	54

The published XML for the minimal demonstration item:

```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 256 256" fill="currentColor">
  <path d="M216,40H40A16,16,0,0,0,24,56V200a16,16,0,0,0,16,16H216a16,16,0,0,0,
16-16V56A16,16,0,0,0,216,40ZM40,56H80V200H40ZM216,200H96V56H216V200Z"/>
</svg>
```

corresponds to the ground-truth string:

```
1decaf83  gt_d      135 chars   21 commands   3 subpaths
M216,40H40A16,16,0,0,0,24,56V200a16,16,0,0,0,16,16H216a16,16,0,0,0,16-16V56A16,16,0,0,0,216,40ZM40
    ,56H80V200H40ZM216,200H96V56H216V200Z
```

Ground truth strings match the upstream source character-for-character. Feeding the ground truth directly into the scoring pipeline yields $S(d, d) = 1.000000$ across all items (§6.3), confirming that evaluation introduces zero distortion.

Geometric Invariance under Non-Affine Deformations. Path grammars comprise linear segments (`M`, `L`, `H`, `V`), polynomial curves (`Q`, `T`, `C`, `S`), and elliptical arcs (`A`). Smooth non-affine spatial transformations map linear segments to sinusoidal trajectories and circular arcs to transcendental curves unsupported by finite Bézier bases. Representing deformed geometry parametrically requires dense polyline sampling, expanding command counts by orders of magnitude. This representation expansion shifts budget constraints and violates operator parsimony. Consequently, the clean evaluation suite preserves unmodified artist paths.

Geometric Reconstruction via Parser Processing. Predicted strings undergo deterministic parsing:

- **Command budgeting.** Budgets are assessed on emitted path syntax: each command letter consumes one unit. Cubic segments incur identical cost to linear segments, rewarding compact parametric curves.
- **Sampling discretization.** Elliptical arcs partition into at most eight sub-segments spanning at most 90°. Polynomial curves sample at fixed rate `CURVE_SAMPLES` = 24. Fixed discretization prevents version drift from altering scores.
- **Coordinate framing.** Raw paths evaluate in $[0, 256]^2$. Viewport specifications map to canvas boundaries via uniform centered scaling.
- **Stroke expansion.** Declared strokes dilate along trajectories via 16-gon circular kernels, unioning with filled interiors under W3C compositing rules.
- **Paint interpretation modes.** Elements declaring stroke attributes without fill are evaluated under both literal fill defaults and outline interpretations, reporting both metrics explicitly.

Table 8 lists model generations and resulting geometric entities.

Table 8: Parser reconstructions across representative model predictions. Command counts determine budget compliance; vertex totals define discrete geometry evaluated by metric functionals.

Model Architecture	Item ID	Characters	Commands	Subpaths	Vertices	Ink Area (px)
`claude-sonnet-4-5`	`20a8ff8d`	197	3	1	3	6,110
`gemini-3.7-flash`	`01001100`	227	8	2	54	9,236
`gpt-5.6-luna`	`1decaf83`	244	12	2	104	13,376

4.4 Complexity Stratification

Icon difficulty varies substantially across the asset pool; evaluating models over unstratified mixtures produces rank instability [5]. I therefore stratify the corpus prior to evaluation along two objective geometric dimensions:

- c_1: Drawing command count k_{gt} (geometric complexity).
- c_2: Topological invariants (β_0, β_1), measuring connected component count β_0 and hole count β_1 via morphological label analysis and Euler characteristics (§6.6).

Stratum dimension c_1 partitions the corpus into command terciles: Simple (S), Medium (M), and Complex (C). Dimension c_2 defines topological classes: single ($\beta_0 = 1, \beta_1 = 0$), holed ($\beta_0 = 1, \beta_1 \geq 1$), and multi ($\beta_0 \geq 2$). Strata derive deterministically from ground-truth geometry, guaranteeing reproducibility.

4.5 Three-Way Corpus Partitioning and Identification Logic

The evaluation corpus partitions into three distinct splits to isolate parametric memorization:

Table 9: Evaluation splits in ToySVG. The procedural disjoint split supplies the non-parametric reference anchor for Proposition 19.

Partition Split	Allocation	Construction Methodology
`ID-CLEAN`	Primary	Canonical Phosphor icons rendered blindly without geometric perturbation
`ID-DEGRADED`	Calibration	Canonical icons downsampled to 128×128 and upscaled with Gaussian blur and noise
`OOD-DISJOINT`	Out-of-Domain	Procedural synthetic paths (Bézier splines, nested loops) absent from public icon datasets

Partitioning operates by evaluating cryptographic hash functions over item identifiers into fixed probability intervals (0.50, 0.20, 0.20, 0.10). Preserving bin boundaries prevents partition drift across iterations. Under Proposition 19, parametric memorization inflates performance on `ID-CLEAN` while providing zero transfer to `OOD-DISJOINT`. The performance difference across splits isolates parametric retrieval from genuine perceptual synthesis.

4.6 Sampling Protocol and Stratum Balance

Sampling draws items within strata to reflect corpus-wide complexity distributions, targeting balanced cell allocations across complexity tiers. Each item records: split assignment, complexity stratum, topological class, ground-truth command and vertex counts, ink area, median stroke thickness, PNG hash, and source asset SHA-256 hash. Sampling derives from a fixed random seed documented in the repository manifest.

4.7 Corpus Hygiene and Security Discipline

- Asset filenames evaluate as `blake2b(seed || asset_id)` truncated to 16 hexadecimal characters. Tasks execute in randomized order, omitting semantic names and category tags from model visibility.
- Image inputs contain zero EXIF headers, text chunks, or embedded comments.
- Hash manifests mapping identifiers to source assets reside outside model execution sandboxes.
- Upstream releases are archived with license files and SHA-256 manifests, ensuring verifiable provenance.

5 Measurement pipeline and metrics

5.1 Pipeline Architecture and Operational Flow

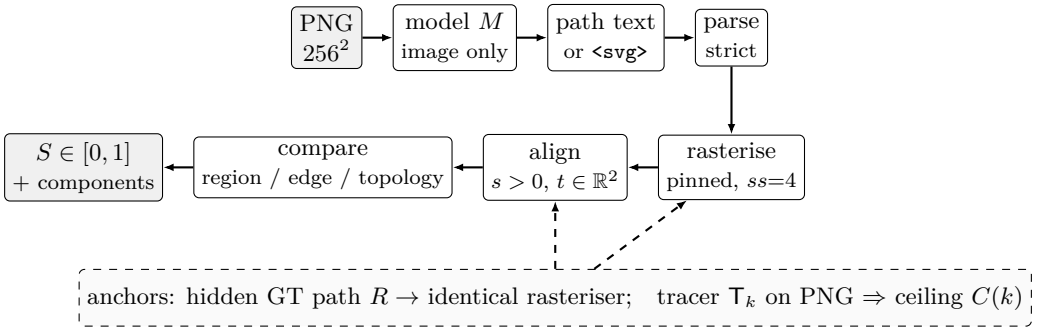


Figure 1: Operational evaluation pipeline. Components following model generation execute deterministically within the repository; ground truth and algorithmic tracers process through the identical rasteriser to guarantee commensurable metrics.

5.2 Deterministic Rasterisation Pipeline

Relying on external SVG renderers introduces metric sensitivity to third-party antialiasing heuristics, discretization algorithms, and library version drift. I therefore implement a pinned, self-contained rasteriser: parametric Bézier curves flatten at fixed, predetermined sample densities (omitting adaptive tolerances that vary across software releases), and fractional pixel coverage evaluates via point sampling on an integer sub-pixel grid of supersampling factor $ss = 4$, followed by uniform box filtering. Scanline spans enforce winding rules directly, resolving interior holes correctly under both `nonzero` and `evenodd` conventions; correctness proofs are provided in Appendix A.

The rasterisation step yields a continuous coverage field $A \in [0, 1]^{n \times n}$ rather than a binary mask. Binary thresholding causes sub-pixel coordinate shifts to trigger discontinuous step changes in overlap; continuous coverage degrades smoothly under minute spatial perturbations, preventing region metrics from penalizing sub-pixel discretization noise. Quantization to discrete multiples of $1/ss^2 = 1/16$ represents an unavoidable finite discretization, whose formal error bounds are established in §6.2.

5.3 Spatial Alignment and Similarity Transformations

When a model reproduces target geometry accurately but shifts its global placement or scale, the output reflects successful visual perception coupled with coordinate displacement. To isolate structural shape perception from spatial calibration, the pipeline optimizes translation and uniform scale prior to metric evaluation. Rotation and reflection remain fixed: on icon corpora, directional orientation encodes essential semantics, and fitting rotational alignments would allow memorized or misoriented symbols to receive unearned credit.

Optimal spatial alignment minimizes loss function $L(s, t)$ over uniform scale $s > 0$ and translation vector $t \in \mathbb{R}^2$:

$$L(s,t) \;=\; \frac{1}{N}\sum_{i=1}^{N}\left|\varphi\left(t + \bar{g} + s\,(p_i - \bar{p})\right)\right| \;+\; \lambda\left|\log\left(s^2\, r\right)\right|, \tag{1}$$

where p_i denote sampled boundary vertices on the predicted path, $\bar{p}$ represents their spatial centroid, $\bar{g}$ denotes the ground-truth centroid, $r = \text{area}(P)/\,\text{area}(R)$ denotes the unaligned area ratio, and penalty coefficient is fixed at $\lambda = 2$. The primary summation penalizes Euclidean boundary divergence via the ground-truth signed distance field φ; the logarithmic regularizer penalizes scale deviations symmetrically, preventing trivial degenerate collapse to isolated boundary points where φ vanishes. Structural convergence and residual bounds are derived in §6.8.

5.4 Component Metric Formulations

Let A and B denote ground-truth and aligned-prediction coverage fields on canvas Ω.

Region Overlap. Soft intersection-over-union evaluates continuous coverage overlap:

$$s_R \;=\; \mathrm{sIoU}(A,B) \;=\; \frac{\sum_{x\in\Omega}\min(A(x),B(x))}{\sum_{x\in\Omega}\max(A(x),B(x))}, \tag{2}$$

with the identity definition $\mathrm{sIoU}(0,0)=1$ when both fields vanish uniformly.

Boundary Alignment. Let ∂A and ∂B represent boundary pixel coordinates extracted from thresholded coverage fields. The symmetric Chamfer distance evaluates:

$$\mathrm{CD}(A,B) \;=\; \tfrac{1}{2}\Big(\frac{1}{|\partial B|}\sum_{y\in\partial B} d(y,\partial A) \;+\; \frac{1}{|\partial A|}\sum_{x\in\partial A} d(x,\partial B)\Big), \tag{3}$$

expressed in pixel units, mapping to boundary score s_E:

$$s_E \;=\; \exp\!\left(-\frac{\mathrm{CD}(A,B)}{\tau}\right), \qquad \tau = 3 \text{ px}. \tag{4}$$

Under scale parameter $\tau = 3$ px, an average boundary divergence of 1 pixel reduces the metric by approximately 28%, while a 3-pixel divergence reduces it by 63%. Diagnostic reports also compute boundary F_1 scores at 2-pixel tolerances and symmetric 95th-percentile Hausdorff distances.

Topological Invariants. Evaluating connected components β_0 and interior hole counts β_1 via 8-connected morphological labeling yields topological error e_T:

$$e_T \;=\; |\beta_0(A)-\beta_0(B)| + |\beta_1(A)-\beta_1(B)|, \qquad s_T \;=\; \frac{1}{1+e_T}. \tag{5}$$

This term penalizes degenerate silhouette approximations: filling interior holes in an outline icon achieves moderate region overlap while failing to preserve topological structure.

Composite Metric Formulation. With fixed weight vector $\mathbf{w} = (0.55, 0.35, 0.10)$, composite score S evaluates:

$$S \;=\; w_R\, s_R + w_E\, s_E + w_T\, s_T. \tag{6}$$

Weights reflect an operational pre-registration preference: region overlap provides robust base alignment, edge Chamfer distance penalizes localized boundary deviations, and topological consistency provides an integer regularizer against structural omission (§6.4).

5.5 Geometric Behavior on Synthetic Reference Pairs

Empirical validation on controlled synthetic geometries clarifies the response characteristics of each metric independently of model behavior. Every panel in this subsection evaluates deterministic shapes whose exact analytical solutions are established a priori via `python -m tools.make_metric_primers`.

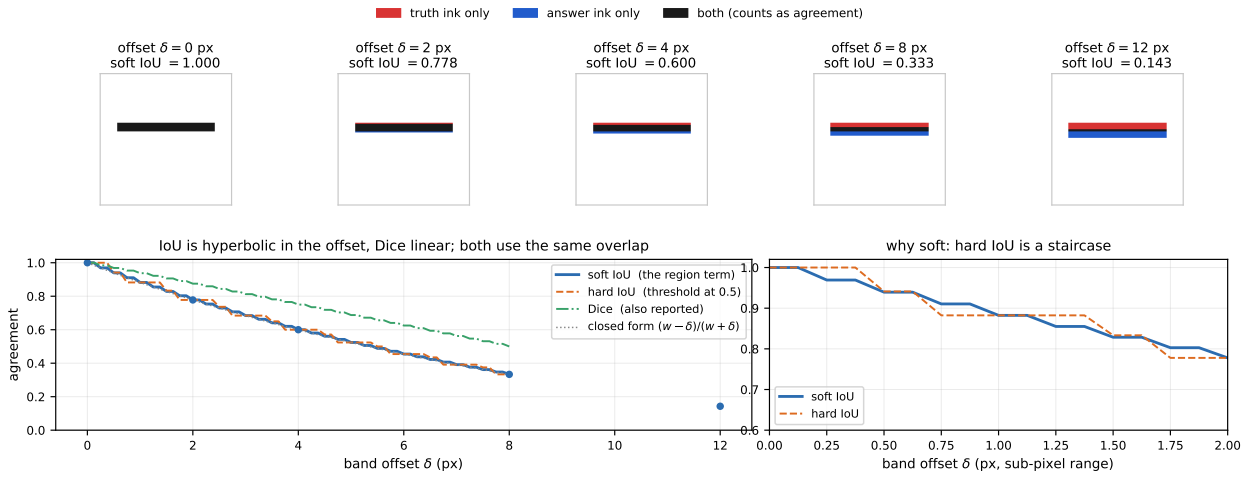


Figure 2: Region metric response under controlled translation. Two parallel rectangular bands of corpus stroke width ($w = 16\,\text{px}$) undergo offset displacement δ. **Top row:** Overlap decomposition; red indicates exclusive ground-truth ink, blue indicates exclusive prediction ink, and black indicates shared overlap summing to $\sum\min$ in Equation 2. **Bottom left:** Measured agreement against analytical forms; soft IoU, hard IoU, and $(w-\delta)/(w+\delta)$ coincide exactly. **Bottom right:** Sub-pixel displacement sweep; hard IoU exhibits discrete 16-step quantization artifacts from grid thresholding, whereas soft IoU remains strictly monotone and continuous.

Figure 2 confirms that measured soft IoU matches analytical closed-form predictions within 10^{-5} across all tested offsets. Furthermore, the area ratio remains identically 1.0000 throughout the displacement sweep because ink mass is conserved, even as overlap drops to 0.143. This discrepancy demonstrates that scalar area measurements fail to capture geometric fidelity.

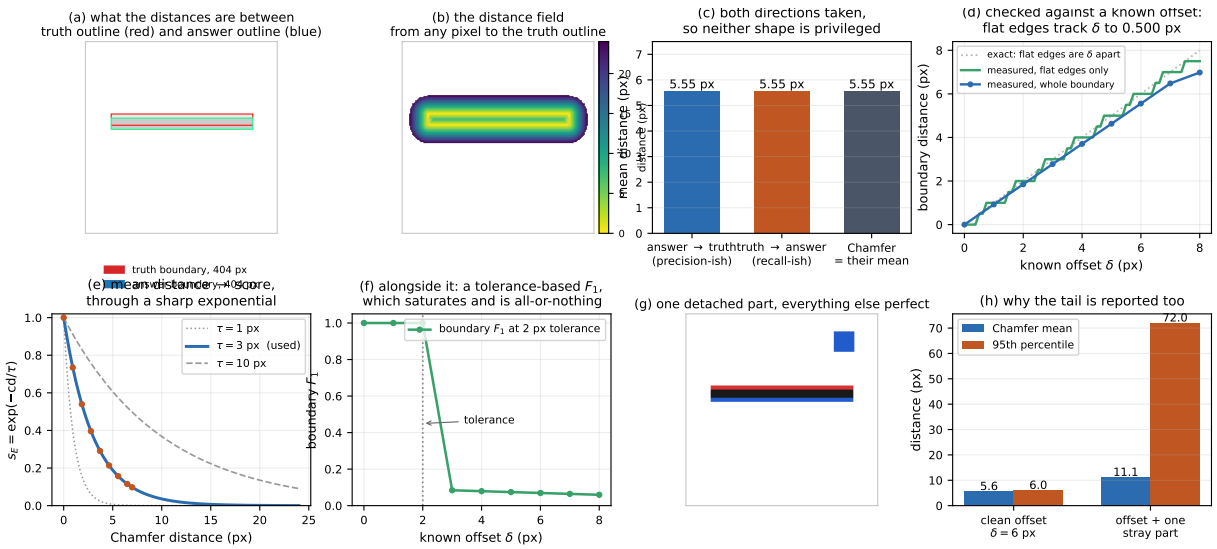


Figure 3: Boundary metric response on parallel 16 px bands offset by $\delta = 6\,\text{px}$. **(a)** Distance evaluations operate over boundary pixel manifolds (red and blue curves, 404 px perimeter) rather than spline control points. **(b)** Euclidean distance field relative to ground-truth boundary. **(c)** Symmetric directed means evaluate to 5.55 px, yielding $s_E = \exp(-5.55/3) = 0.157$. **(d)** Controlled offset sweep; flat-edge measurements track true displacement within 0.500 px discretization limits. **(e)** Exponential decay curves across decay scales $\tau \in \{1, 3, 10\}$. **(f)** Binary boundary F_1 demonstrates all-or-nothing threshold saturation. **(g),(h)** Outlier sensitivity under localized part displacement, where 95th-percentile Hausdorff shifts from 6.0 px to 72.0 px.

Figure 3 establishes two properties: the metric is mathematically symmetric across argument order, and 95th-percentile Hausdorff metrics effectively register isolated structural outliers that leave mean Chamfer distances moderately perturbed.

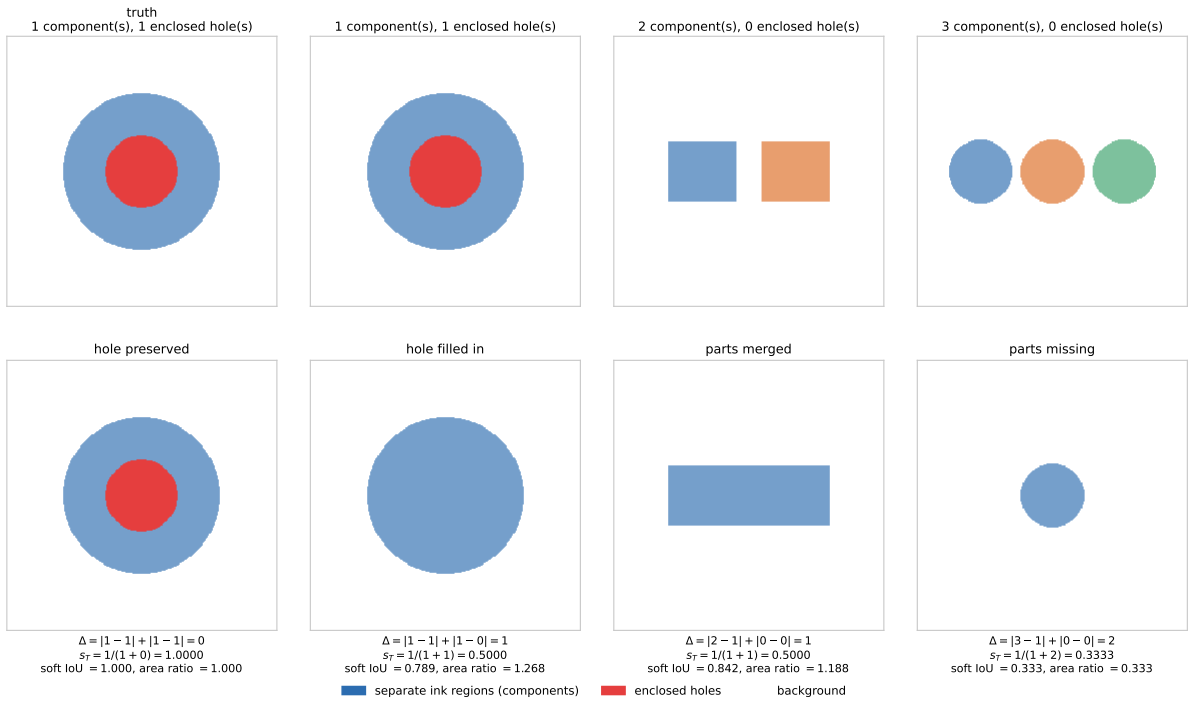


Figure 4: Topological metric behavior across controlled structural variations. Distinct components are assigned unique colors; enclosed holes are highlighted in red. **Left to right:** Exact topology preservation ($\Delta = 0$, $s_T = 1.0000$, sIoU $= 1.000$); hole filled into solid silhouette ($\Delta = 1$, $s_T = 0.5000$, sIoU $= 0.789$); two separate components merged into single manifold ($\Delta = 1$, $s_T = 0.5000$, sIoU $= 0.842$); omission of two components ($\Delta = 2$, $s_T = 0.3333$, sIoU $= 0.333$).

Figure 4 illustrates why unblended evaluation is necessary: filling an interior hole maintains high region overlap (sIoU $= 0.789$), yet the topology metric records $s_T = 0.5000$, exposing the loss of structural geometry.

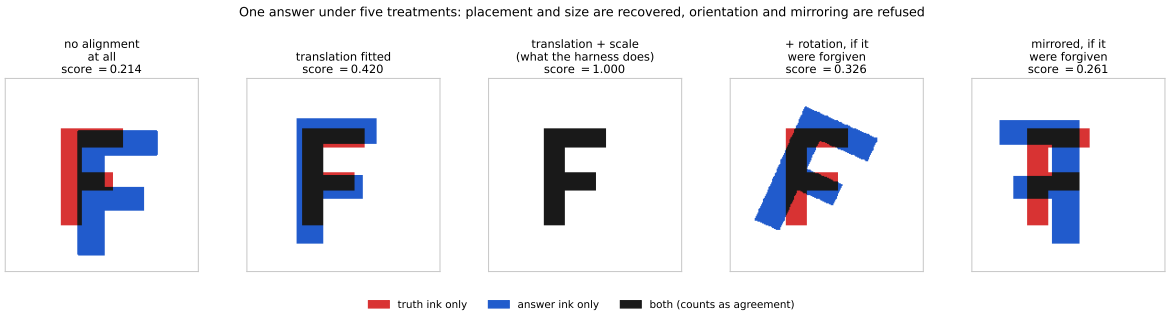


Figure 5: Similarity transform behavior on a chiral test shape displaced by $(34, 20)$ px and scaled by factor 1.28. Raw unaligned overlap yields $S = 0.214$. Translation optimization increases score to 0.420. Combined translation and scale optimization restores $S = 1.0000$, recovering true scale within 10^{-4}. Unpermitted rotation and reflection yield $S = 0.326$ and $S = 0.261$ respectively, confirming that metric penalties correctly reject rotated shapes.

Figure 5 validates Proposition 8: congruent shapes subject to translation and uniform dilation achieve $S = 1.0000$, establishing operational commensurability between model generations and algorithmic tracer ceilings.

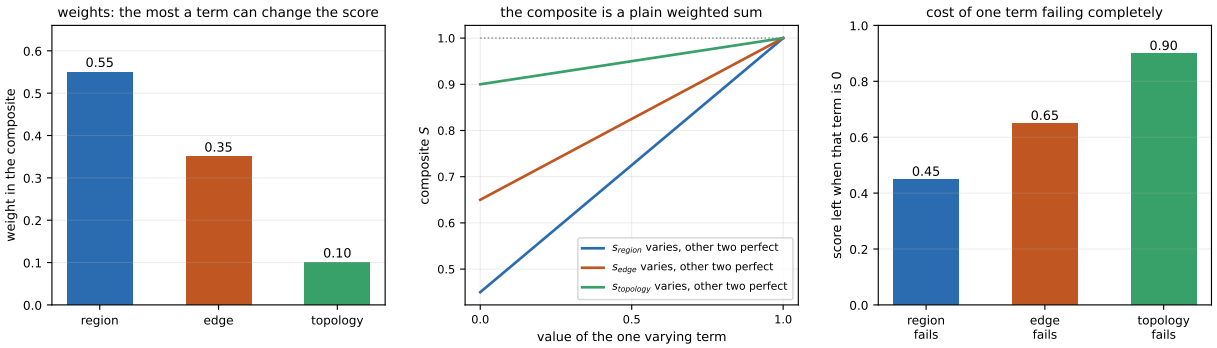


Figure 6: Composite weighting mechanics. **Left:** Linear weighting coefficients. **Middle:** Component response slopes holding complementary metrics at unity. **Right:** Bounded metric floors under individual component failures: total region failure permits maximum score 0.45; boundary failure permits 0.65; topological failure permits 0.90.

Figure 6 establishes the metric interpretation rule: achieving an aggregate score below 0.45 requires concurrent failure across multiple evaluation dimensions.

5.6 Reference Anchors and Normalization Baselines

To establish valid operational scales, every item is evaluated against four reference anchors:

Table 10: Empirical reference anchors. Control baselines verify pipeline calibration; ceilings quantify algorithmic upper bounds.

Anchor Identifier	Classification	Mathematical Definition
`blank`	Floor	Null string emitting zero ink; establishes empty generation floor
`bbox`	Floor	Axis-aligned bounding box enclosing ground-truth ink
`ellipse`	Floor	Moment-matched second-order covariance ellipse
`centroid_dot`	Floor	Concentric circular disk located at spatial centroid $\bar{g}$
`sanity_shift`	Control	Ground truth translated by $(30, 0)$ px; must evaluate to $S \approx 1.000$ after alignment
`trace`	Ceiling	Output of deterministic tracer T_k executed on input PNG at budget k

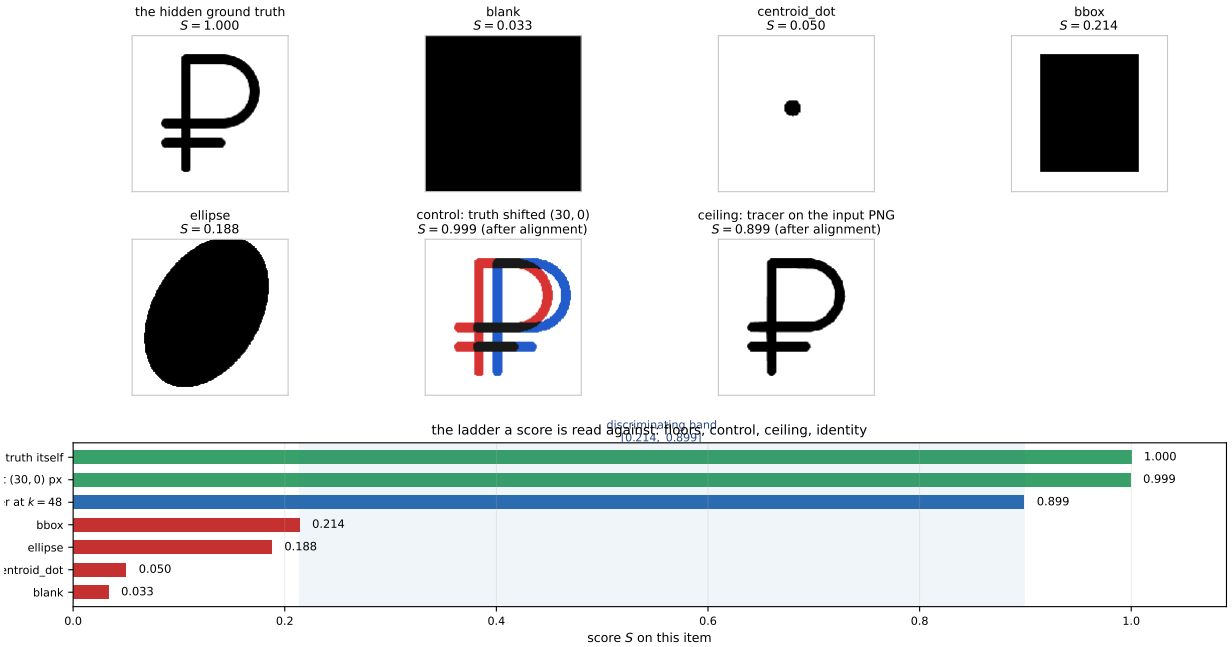


Figure 7: Metric calibration ladder on demonstration item `01001100`. **Top:** Visual geometry across floor anchors, control perturbation, and tracer ceiling. **Bottom:** Ordered scores: null blank (0.033), centroid dot (0.050), matched ellipse (0.188), bounding box (0.214), procedural tracer $C(48) = 0.899$, translation control (0.999), and ground-truth identity (1.000).

The reference ceiling defines the parsimony ratio $\rho(k)$:

$$\rho(k) \;=\; \frac{S(M,k)}{C(k)}, \tag{7}$$

which normalizes raw performance against procedural tracing efficiency. A parsimony ratio $\rho(k) > 1.0$ indicates that the model achieves higher visual fidelity than the procedural tracer using equivalent or fewer command primitives.

5.7 Statistical Estimation and Resampling Protocol

Item scores $S_i \in [0,1]$ exhibit non-Gaussian, multi-modal distributions. Consequently, all reported confidence intervals derive from stratified bootstrap resampling ($B = 10^4$ replications within stratum cells), reporting 2.5th and 97.5th percentiles. Comparative hypothesis testing between models evaluates paired per-item differences $d_i = S(M_1, x_i) - S(M_2, x_i)$ via paired permutation tests.

5.8 Empirical Worked Example on Benchmark Items

To demonstrate pipeline mechanics, this subsection traces three empirical items evaluated under budget tier B1 ($k = 48$) across Simple, Medium, and Complex strata within the clean evaluation partition.

5.9 Detailed Numerical Walkthrough

Evaluating item `01001100` clarifies the diagnostic role of unblended components:

Model Visual Stimulus. The model receives prompt template `p1` alongside input image Figure 8a and explicit constraint $k = 48$, without asset metadata or category names.

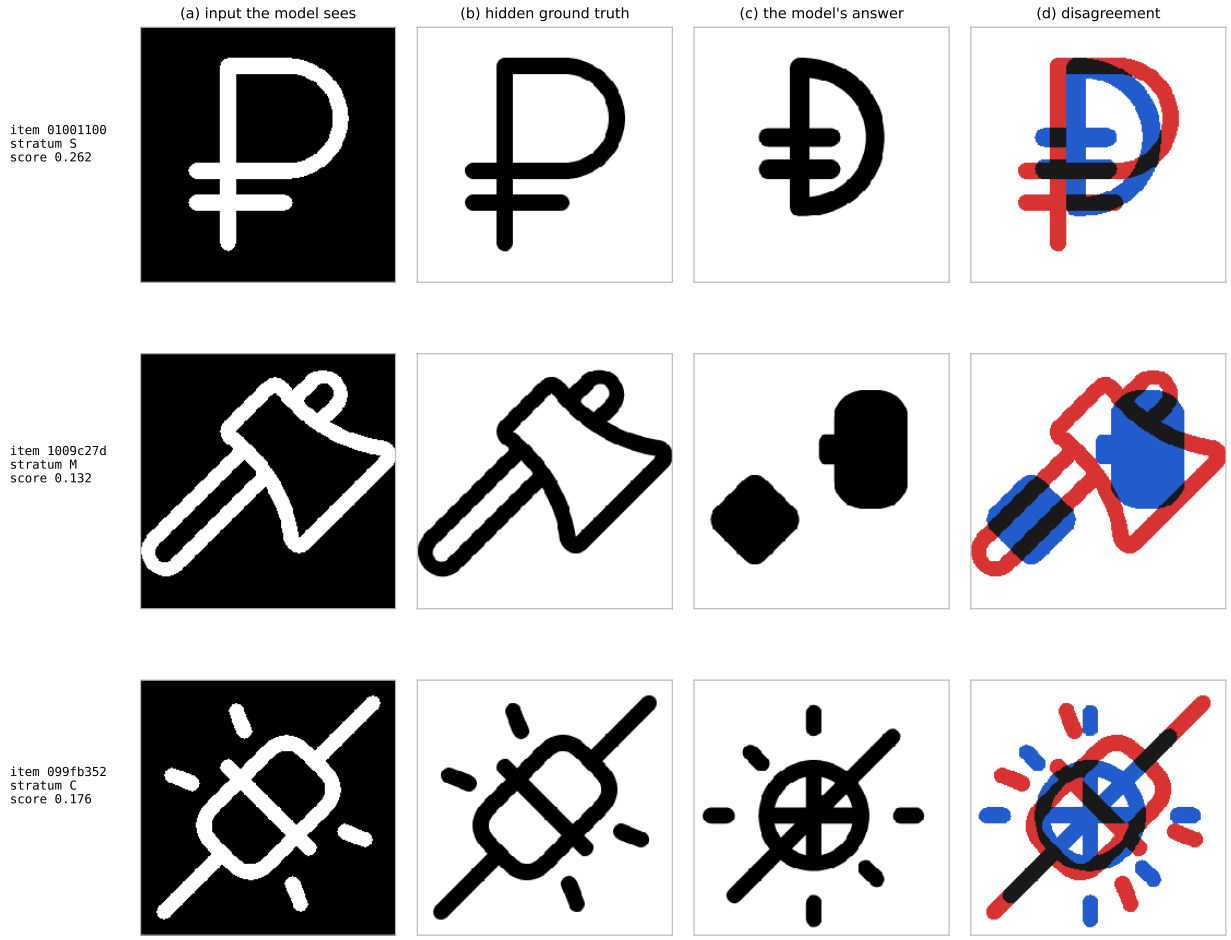


Figure 8: Pipeline execution on three test instances. **(a)** Input PNG images presented to model. **(b)** Hidden ground-truth vector geometry. **(c)** Predicted paths parsed and aligned by pipeline. **(d)** Geometric discrepancy: red indicates omitted ground-truth ink, blue indicates extraneous prediction ink, and black indicates spatial overlap.

Model Output Syntax. The model emits nine commands, staying within budget:

```
<svg viewBox="0 0 256 256"><path d="M80 40V216M40 128H120M40 168H120
M80 40C140 40 176 80 176 128C176 176 140 216 80 216"
fill="none" stroke="currentColor" stroke-width="24"
stroke-linecap="round" stroke-linejoin="round"/></svg>
```

The output constructs a vertical spine, two horizontal crossbars, and an enclosed curved loop, selecting stroke-based primitives that match ground-truth topology.

Geometric Pipeline Processing. The parser expands the 24-unit stroke into planar geometry via disk sweeping, verifies boundary containment, confirms $\beta_0 = 1$ and $\beta_1 = 1$, and fits similarity parameters (s^*, t^*).

Error Localization. Fitted scale evaluates to $s^* = 0.79$, with mean boundary divergence of 8.2 px and 95th-percentile Hausdorff distance of 29 px. The loop feature is drawn narrower than ground truth, reducing continuous overlap to $s_R = 0.254$. Topological fidelity evaluates to $s_T = 1.000$, yielding composite score $S = 0.262$.

5.10 Score Computation: Explicit Evaluation Trace

All reported scores compute deterministically via `python -m tools.explain_score`:

Table 11: Decomposed performance metrics on three representative model predictions. Emitted command counts confirm that generation was not artificially constrained by budget limits.

Item ID	Stratum	Commands	s_R	s_E	s_T	S	$C(48)$	Primary Structural Error
`01001100`	S	9	0.254	0.065	1.000	0.262	0.899	Correct topology; distorted aspect proportions
`1009c27d`	M	25	0.191	0.020	0.200	0.132	0.889	Outline rendered as solid silhouette
`099fb352`	C	26	0.266	0.043	0.143	0.176	0.482	Semantically plausible alternative geometry
Mean		20	0.237	0.043	0.448	0.190	0.757	

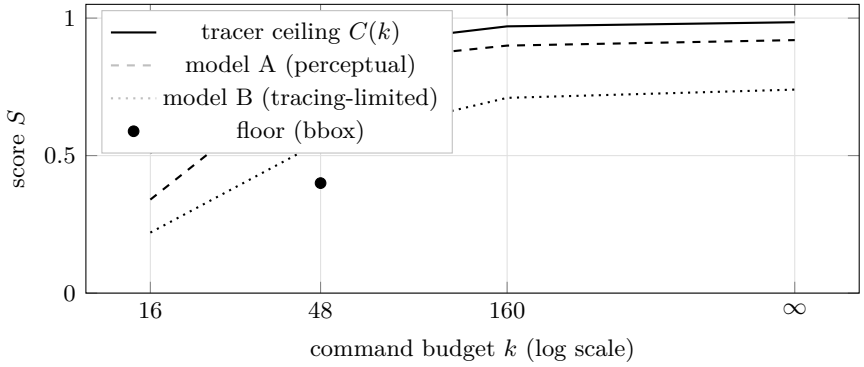


Figure 9: Theoretical budget curves across evaluation tiers, illustrating performance regimes relative to tracer ceilings and bounding-box baselines.

Five Pipeline Execution Stages. Execution proceeds strictly through five successive operations:

1. **Syntax parsing.** Count command tokens, validate XML structure, and apply Minkowski dilation to stroked paths.
2. **Continuous rasterisation.** Sample geometry onto a 256×256 grid at $ss = 4$ supersampling to obtain coverage field $A \in [0,1]^{256\times256}$.
3. **Spatial alignment.** Solve optimization problem (1) to estimate scale s^* and translation t^*.
4. **Metric evaluation.** Compute region overlap s_R, boundary score s_E, and topological score s_T.
5. **Anchor calibration.** Evaluate reference floors and procedural tracer ceiling $C(k)$.

Component Formulation and Metric Definitions. Let G and P denote discrete coverage arrays:

$$s_R = \frac{\sum_{u,v} \min\{G(u,v),\, P(u,v)\}}{\sum_{u,v} \max\{G(u,v),\, P(u,v)\}}, \tag{8}$$

$$s_E = \exp\!\left(-\frac{\mathrm{cd}}{\tau}\right), \qquad \tau = 3 \text{ px}, \qquad \mathrm{cd} = \tfrac{1}{2}\Big(\overline{d(P\to G)} + \overline{d(G\to P)}\Big), \tag{9}$$

$$s_T = \frac{1}{1+\Delta}, \qquad \Delta = |\Delta\beta_0| + |\Delta\beta_1|, \tag{10}$$

$$S = w_R s_R + w_E s_E + w_T s_T, \qquad (w_R, w_E, w_T) = (0.55,\, 0.35,\, 0.10). \tag{11}$$

Step-by-Step Numerical Evaluation Trace.

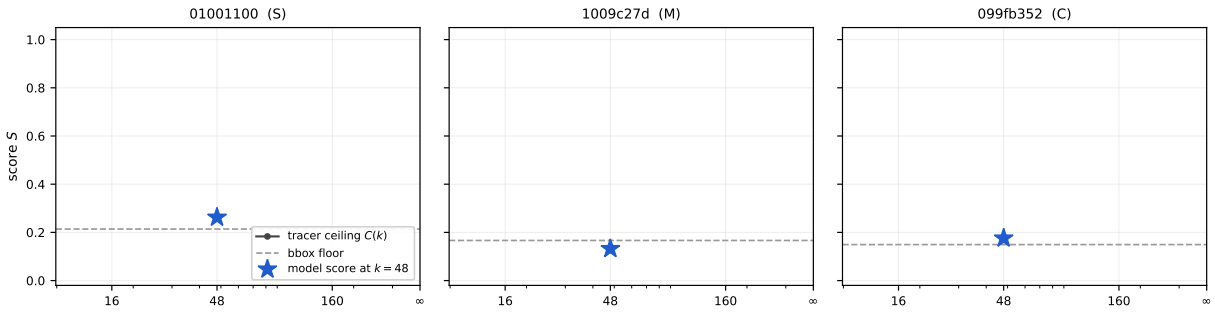


Figure 10: Empirical tracer ceilings evaluated on demonstration items across budget tiers $k \in \{16, 48, 160, \infty\}$. Dashed lines indicate bounding-box floors; stars mark model predictions at $k = 48$.

Table 12: Intermediate evaluation trace for item `01001100`, detailing exact numerical values per stage.

Stage	Metric Evaluation and Numerical Values
1. Parse	9 drawing commands (≤ 48), 4 subpaths, single element, `fill="none"`, `stroke-width="24"`
2. Render	Ground-truth ink area: 9,146.2 px; pre-alignment prediction ink: 14,723.0 px
3. Align	Scale $s^* = 0.7897$, translation $t^* = (+18.68, -18.05)$ px; aligned ink: 9,187.2 px; area ratio 1.004
4a. Region	$\sum \min = 3{,}711.1$, $\sum \max = 14{,}622.3 \implies s_R = 3{,}711.1/14{,}622.3 = 0.2538$
4b. Edge	$\overline{d(P \to G)} = 7.285$ px, $\overline{d(G \to P)} = 9.109$ px $\implies$ cd $= 8.197$ px; $s_E = e^{-8.197/3} = 0.0651$
4c. Topology	Ground truth: $(1, 1)$; prediction: $(1, 1) \implies \Delta = 0$; $s_T = 1/(1+0) = 1.0000$
5. Composite	$0.55(0.2538) + 0.35(0.0651) + 0.10(1.0000) = 0.13959 + 0.02277 + 0.10000 =$ **0.26236**
Anchors	Identity 1.000; bounding box 0.214; ellipse 0.188; blank 0.033; tracer $C(48) = 0.899$; parsimony ratio $\rho = 0.292$

Geometric Analysis of Overlap Metrics. For two rectangular bands of stroke width w offset by transverse distance δ, analytical IoU follows:

$$\text{IoU} = \frac{w - \delta}{w + \delta}. \tag{12}$$

For stroke thickness $w = 16$ px and offset $\delta = 8$ px, theoretical overlap evaluates to IoU $= 8/24 \approx 0.333$. Thin vector paths exhibit substantial sensitivity to minor spatial deviations, explaining why raw overlap scores appear modest even when visual structures correspond closely.

Ground-Truth Complexity and Budget Allocation. Ground-truth icons in vector datasets are not guaranteed to be parsimonious: five items in the test pack require more than 48 commands in their published form. Evaluating such paths against budget $k = 48$ triggers over-budget validity gates. This property justifies reporting parsimony ratio $\rho(k)$: normalising by tracer ceiling $C(k)$ evaluates models against what is computationally achievable within the prescribed budget.

Independent Arithmetic Reproduction. All stage metrics reproduce via command execution:

```
python -m tools.explain_score --pack data/pack \
    --run data/runs/gemini-2.5-flash-demo --item 01001100afd9128f
```

5.11 Metric Behavior on Empirical Model Generations

Evaluating model predictions on empirical items illustrates metric sensitivity:

Figure 12 confirms that on item `1009c27d`, rendering an outline icon as a solid silhouette incurs a topological penalty ($\Delta = 4$, $s_T = 0.200$) despite close agreement in total surface area.

Table 13: Intermediate evaluation quantities across the three worked-example items.

Quantity	`01001100`	`1009c27d`	`099fb352`	Operational Interpretation
Commands spent	9	25	26	Compliant with budget tier $k = 48$
$\sum$ min	3711.1	4333.9	5183.8	Continuous intersection coverage
$\sum$ max	14,622.3	22,638.5	19,452.4	Continuous union coverage
s_R	0.2538	0.1914	0.2665	Continuous soft Jaccard ratio
$\overline{d(P \to G)}$	7.285	6.570	9.508	Boundary distance: prediction to truth (px)
$\overline{d(G \to P)}$	9.109	16.967	9.434	Boundary distance: truth to prediction (px)
cd	8.197	11.769	9.471	Symmetric Chamfer mean (px)
s_E	0.0651	0.0198	0.0426	Exponential boundary fidelity
Δ	0	4	6	Topological component and hole discrepancies
s_T	1.0000	0.2000	0.1429	Topological congruence
S	0.2624	0.1322	0.1757	Weighted composite score
$C(48)$	0.899	0.889	0.482	Algorithmic tracer ceiling at $k = 48$
ρ	0.292	0.149	0.365	Parsimony ratio relative to tracer

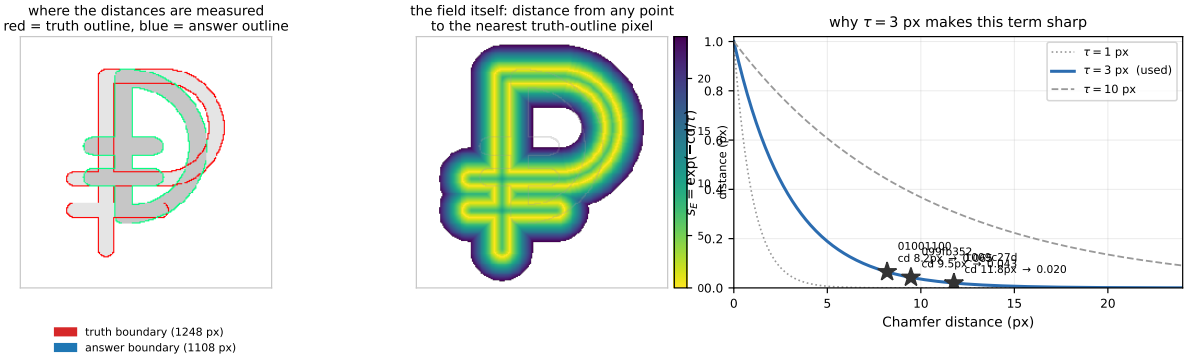


Figure 11: Boundary Chamfer evaluation on demonstration item `01001100`. **Left:** Boundary pixel contours (truth in red, prediction in blue). **Middle:** Distance transform field from ground-truth boundary. **Right:** Boundary metric decay across scale parameters $\tau \in \{3, 5, 10\}$.

5.12 Empirical Kaggle Evaluation: Multi-Model Benchmark

The benchmark specification was deployed as an automated Kaggle Benchmarks task, evaluating the 32-item test pack at budget tier B1 ($k = 48$) across seven frontier vision-language models without item substitution. Evaluation routines match local implementations bit-for-bit, verified in §5.12.

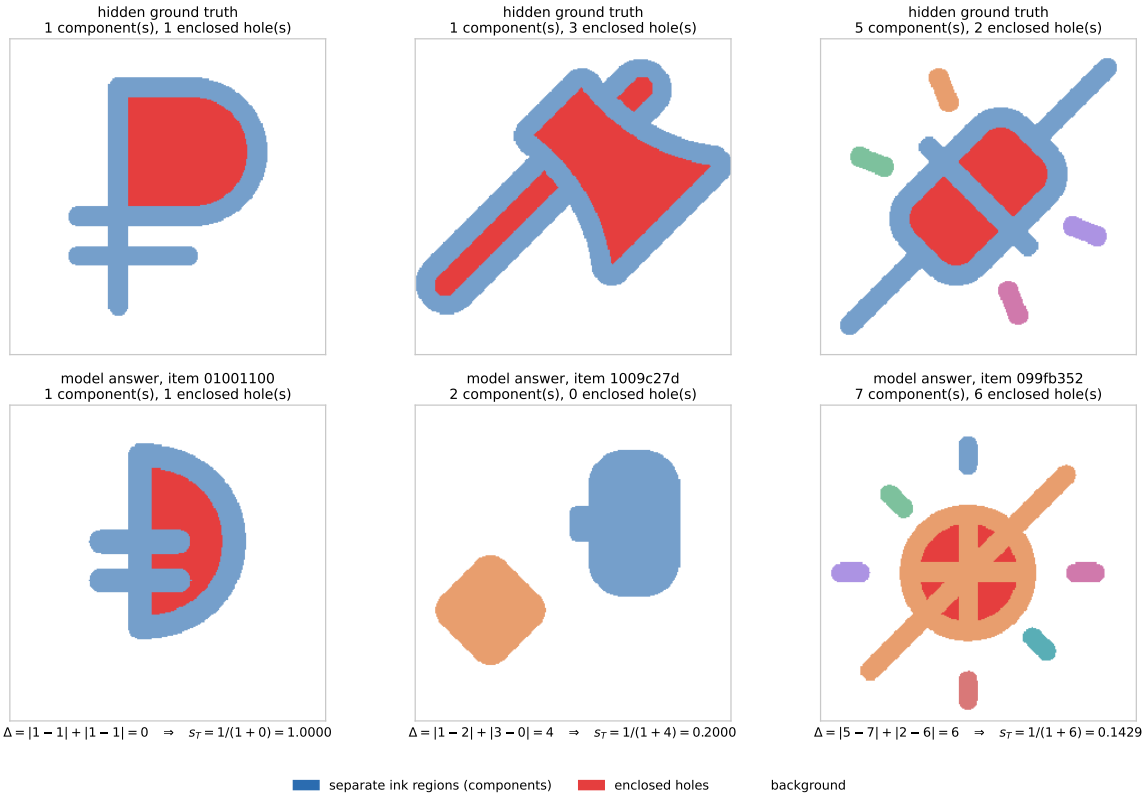


Figure 12: Topological component parsing across demonstration items. Components receive unique color labels; interior holes are highlighted in red. **Top row:** Ground-truth paths. **Bottom row:** Model outputs, displaying topological error $\Delta = |\Delta\beta_0| + |\Delta\beta_1|$ and metric s_T.

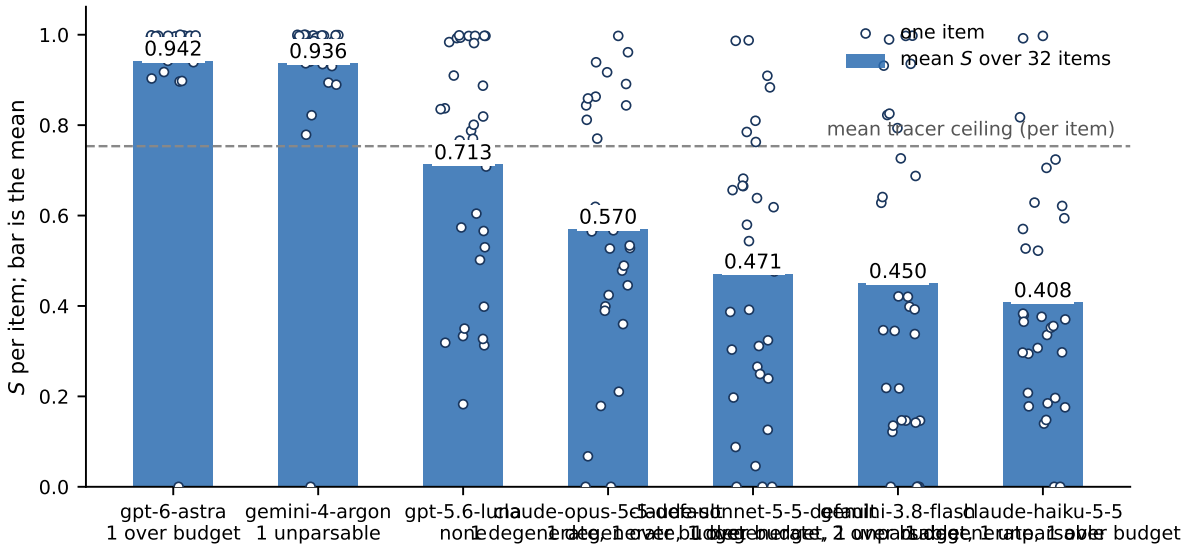


Figure 13: Per-item model score distributions on the 32-item Kaggle evaluation. Circles indicate individual item scores S_i; vertical bars indicate aggregate means; the dashed horizontal line marks mean tracer ceiling 0.7535. Annotations indicate failure gate activations per model.

Table 14: Kaggle Benchmarks evaluation results across seven frontier multimodal models on the 32-item test pack at budget $k = 48$. The table presents composite geometric score S, parsimony ratio ρ, tracer ceiling comparisons, and gate distributions.

Model	mean S	best	worst	mean ρ	above ceiling	Gates that fired
`gpt-6-astra`	0.9419	1.000	0.000	1.291	30/32	1 over budget
`gemini-4-argon`	0.9364	1.000	0.000	1.366	30/32	1 unparsable
`gpt-5.6-luna`	0.7126	0.999	0.183	0.967	17/32	none
`claude-opus-5-5-default`	0.5700	0.997	0.000	0.747	7/32	1 degenerate, 1 over budget
`claude-sonnet-5-5-default`	0.4710	0.988	0.000	0.591	3/32	1 degenerate, 1 over budget, 1 unparsable
`gemini-3.8-flash`	0.4499	0.998	0.000	0.612	7/32	1 degenerate, 2 over budget, 1 unparsable
`claude-haiku-5-5`	0.4076	0.997	0.000	0.516	2/32	1 degenerate, 1 over budget

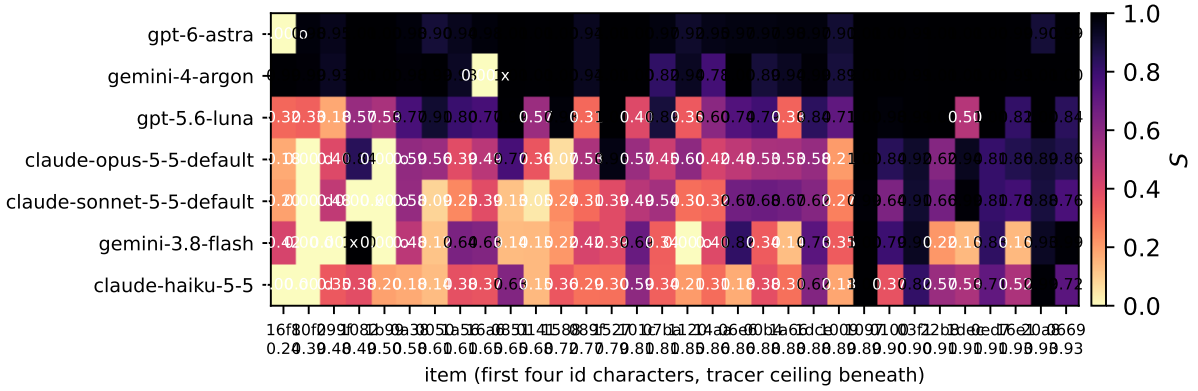


Figure 14: Performance matrix across models and test instances, ordered left-to-right by increasing tracer ceiling (0.2424 to 0.9294). Symbols denote gate triggers: `x` indicates unparsable syntax; `o` indicates command budget violations.

As documented in Table 14, `gpt-6-astra` achieves the highest aggregate mean score 0.9419 with parsimony ratio 1.291, surpassing the deterministic tracer ceiling on 30 of the 32 items, with a single budget violation gate trigger. Close behind, `gemini-4-argon` achieves mean score 0.9364 with parsimony ratio 1.366, also exceeding the tracer ceiling on 30 of 32 items, with one unparsable syntax trigger. `gpt-5.6-luna` achieves mean score 0.7126 with parsimony ratio 0.967, demonstrating flawless syntactic discipline with zero validity gate triggers across the entire pack (32/32 valid) and an item score floor of 0.183. Lower tiers reveal pronounced degradation: `claude-opus-5-5-default` achieves 0.5700, `claude-sonnet-5-5-default` achieves 0.4710, `gemini-3.8-flash` achieves 0.4499, and `claude-haiku-5-5` achieves 0.4076, frequently losing entire items to geometric degeneracy or command budget exhaustion.

Remote Deployment Parity and Environmental Invariance. Tracer ceilings depend exclusively on input rasters and budget constraints. Recomputing ceilings locally from corpus files reproduces remote Kaggle values within 10^{-12} across all 32 items, verifying runtime parity.

Diagnostic Artifact Access. Evaluation runs export complete artifact bundles: raw generation strings, component metric scores, validity flags, fitted affine parameters, and three-panel visual diagnostic strips.

Sample Scope. Current evaluation logs reflect seven frontier multimodal models evaluated across the full 32-item benchmark pack, establishing execution determinism and diagnostic utility on remote cloud infrastructure.

5.13 Qualitative Visual Plate Analysis

Figures 15 and 16 illustrate representative model generations across the evaluated test items alongside input stimuli and ground-truth geometry, organized by tracer ceiling difficulty.

6 Properties and proofs

This section proves foundational mathematical properties of the benchmark, establishes analytical bounds necessary for score interpretation, and derives statistical sample size requirements. Proofs are constructive; where deductions rely on operational assumptions, preconditions are stated explicitly and mapped to empirical verification tests in §8.

6.1 Lipschitz Continuity of Signed Distance Fields

Lemma 3 (Distance functions are 1-Lipschitz)**.** *Let $A \subseteq \mathbb{R}^2$ be a non-empty closed set and define Euclidean distance $d(z, A) = \inf_{a \in A} \|z - a\|$. Then $|d(z, A) - d(w, A)| \leq \|z - w\|$ for all $z, w \in \mathbb{R}^2$.*

Proof. Let $a \in A$ be arbitrary. By the triangle inequality on $\mathbb{R}^2$, $d(z, A) \leq \|z-a\| \leq \|z-w\|+\|w-a\|$. Taking the infimum over $a \in A$ yields $d(z, A) \leq \|z-w\|+d(w, A)$, which implies $d(z, A)-d(w, A) \leq \|z - w\|$. Reversing the roles of z and w yields $d(w, A) - d(z, A) \leq \|z - w\|$, establishing the inequality. □

Proposition 4 (Signed distance fields are 1-Lipschitz)**.** *Let $R \subset \mathbb{R}^2$ be a bounded region with non-empty interior and topological boundary ∂R. Define signed distance field $\varphi : \mathbb{R}^2 \to \mathbb{R}$ by*

$$\varphi(z) \;=\; d(z, \partial R) \cdot (\mathbf{1}[z \notin R] - \mathbf{1}[z \in R]),$$

positive on the exterior and negative on the interior. Then $|\varphi(z)-\varphi(w)| \leq \|z-w\|$ for all $z, w \in \mathbb{R}^2$.

Proof. Three exhaustive configurations arise:

(i) *Both points lie in the exterior R^c:* Here $\varphi = d(\cdot, \partial R)$, and Lemma 3 applies directly.

(ii) *Both points lie in the interior R:* Here $\varphi = -d(\cdot, \partial R)$, and Lemma 3 applies directly.

(iii) *Points lie on opposite sides:* Let $z \in R^c$ and $w \in R$. The line segment $[z, w]$ is connected and intersects both R and R^c. Because R and R^c are closed in their respective subspace topologies, $[z, w] \cap \partial R \neq \emptyset$. Selecting boundary point $q \in [z, w] \cap \partial R$ gives:

$$|\varphi(z) - \varphi(w)| \;=\; d(z, \partial R) + d(w, \partial R) \;\leq\; \|z - q\| + \|q - w\| \;=\; \|z - w\|,$$

because q lies on the straight segment joining z and w. This completes the proof in all cases.

□

Corollary 5 (Lipschitz bounds on alignment objective)**.** *The first term of alignment loss* (1) *is 1-Lipschitz with respect to translation: for all $t, \delta \in \mathbb{R}^2$,*

$$\Big|\frac{1}{N}\sum_{i=1}^{N} |\varphi(t + \bar{g} + s(p_i - \bar{p}))| - \frac{1}{N}\sum_{i=1}^{N} |\varphi(t + \delta + \bar{g} + s(p_i - \bar{p}))|\Big| \;\leq\; \|\delta\|,$$

and $s\bar{\rho}$-Lipschitz with respect to uniform scale s, where $\bar{\rho} = \frac{1}{N}\sum_{i=1}^{N} \|p_i - \bar{p}\|$ denotes the mean radius of sampled prediction points.

Proof. Applying Proposition 4 pointwise in conjunction with inequality $||u| - |v|| \leq |u - v|$ yields the translation bound. For uniform scale, boundary vertex p_i undergoes spatial displacement $\|(s' - s)(p_i - \bar{p})\|$, bounding total variation by $|s' - s|\frac{1}{N}\sum_{i=1}^{N} \|p_i - \bar{p}\|$. □

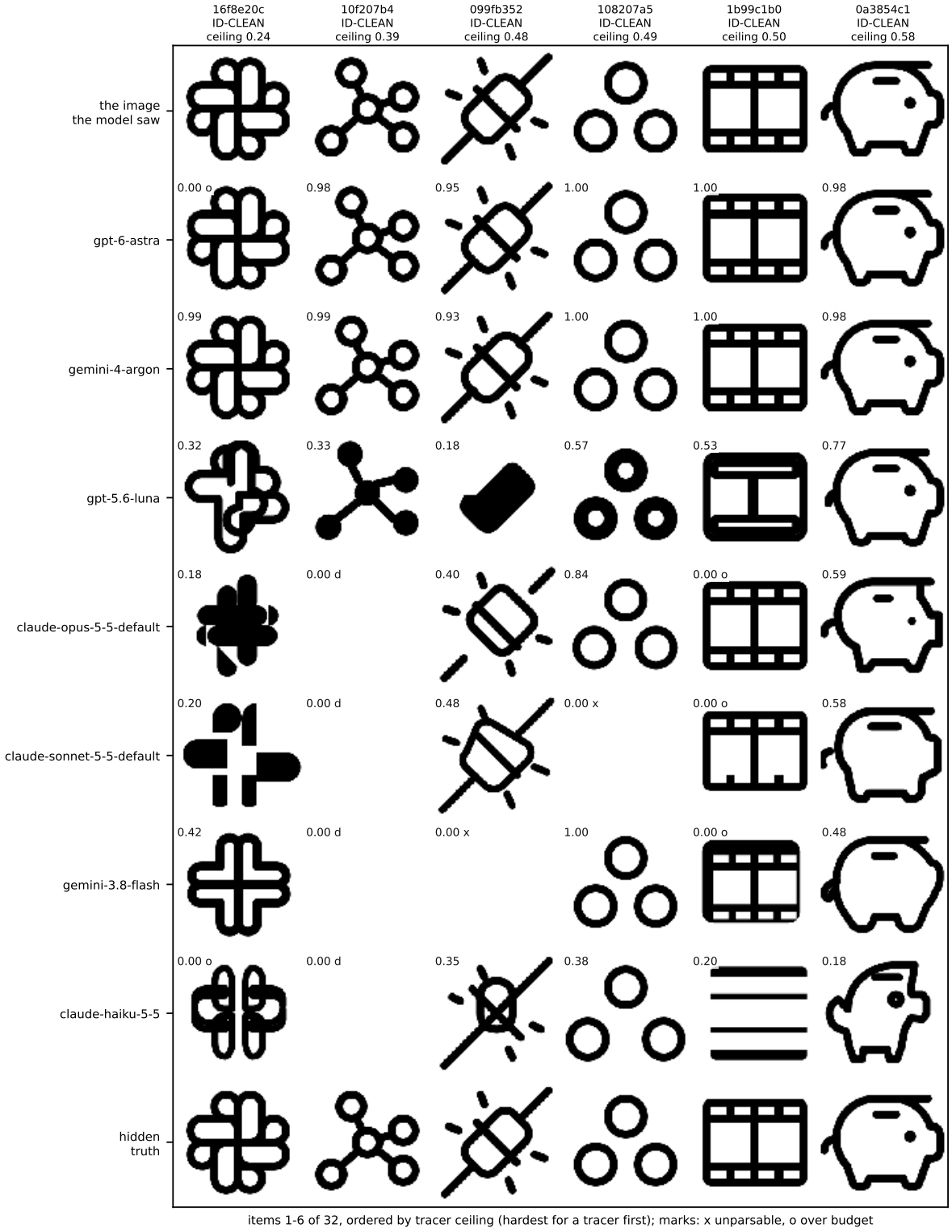


Figure 15: Visual evaluation plate for items 1–6 of 32, ordered by tracer ceiling difficulty. Rows correspond to input images, model generations, and ground-truth vector geometry. Numeric labels report composite score S; `x` marks parsing failures; `o` marks budget violations.

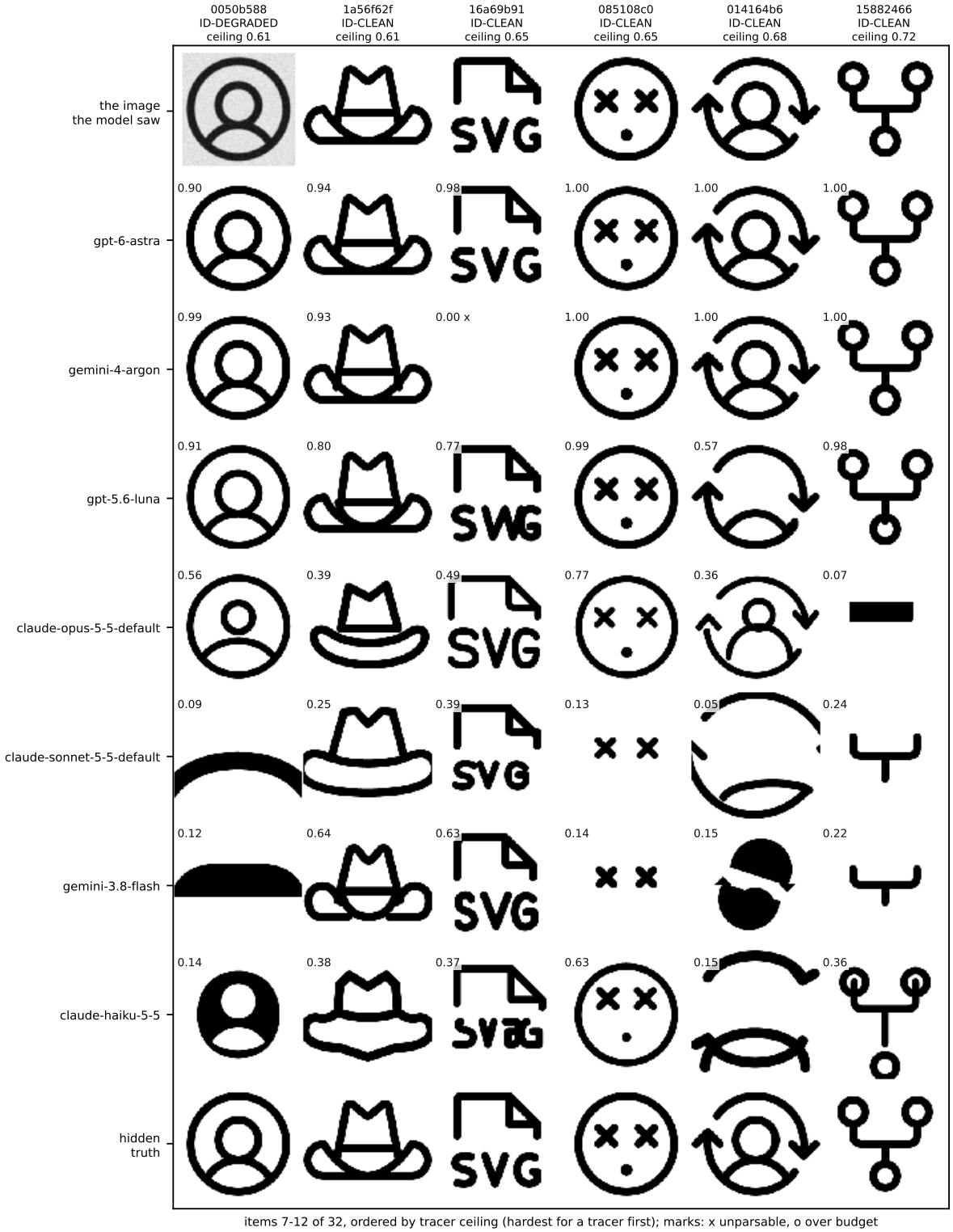


Figure 16: Visual evaluation plate for items 7–12 of 32. Item `20a8ff8d` represents the `ID-DEGRADED` condition, displaying visual artifacts while preserving exact underlying vector geometry.

6.2 Rasterisation Discretization Error Bounds

Lemma 6 (Coverage estimator error bound)**.** *Let $R \subseteq [0,n)^2$ possess rectifiable boundary ∂R of total Euclidean length ℓ in pixel units. Let $\widehat{A}$ denote the $ss \times ss$ supersampled coverage estimator defined in Appendix A. Then:*

(i) $\widehat{A}(x) = A(x)$ exactly for every pixel $x \in \Omega$ whose open interior does not intersect ∂R;

(ii) $\sum_{x\in\Omega} |\widehat{A}(x) - A(x)| \;\le\; \dfrac{2\ell}{ss} + O\!\left(\dfrac{1}{ss^2}\right)$.

Proof. (i) If pixel x does not intersect ∂R, every sub-cell lies entirely inside R or entirely outside R. Point sampling at sub-cell centers classifies containment with zero error, matching the continuous area fraction.

(ii) Non-zero error is confined strictly to pixels intersecting boundary ∂R. Rescaling to sub-pixel cell units, the boundary curve has length $\ell \cdot ss$. A planar curve of length L intersects at most $2L + 1$ unit lattice cells. Thus, at most $2\ell\, ss + 1$ sub-cells are traversed by ∂R. Each traversed sub-cell contributes at most $1/ss^2$ to the total area discrepancy, bounding aggregate error by $(2\ell\, ss + 1)/ss^2 = 2\ell/ss + 1/ss^2$. □

Remark 7 (Common-mode cancellation). Bound $O(\ell/ss)$ represents an adversarial worst-case upper limit. Under $ss = 4$ and typical icon boundary lengths $\ell \approx 1{,}000$ px, aggregate error remains under 1% of canvas area. Crucially, the identical lattice rasteriser evaluates both ground truth and prediction, inducing strongly correlated quantization errors that cancel in soft IoU ratios.

6.3 Identity Exactness and Metric Reflexivity

Proposition 8 (Identical geometry scores exactly unity)**.** *If predicted path geometry matches ground truth identically, the evaluation functional satisfies $S(d,d) = 1$ with all three component metrics evaluating to unity.*

Proof. The pinned rasteriser is a deterministic pure function of geometry: scanline sorting is stable and accumulator operations are order-invariant. Identical geometry yields bit-identical coverage arrays $B = A$. Evaluating soft intersection-over-union yields $\mathrm{sIoU}(A, A) = 1$. Boundary manifolds coincide identically, giving Chamfer distance $\mathrm{CD} = 0$ and $s_E = \exp(0) = 1$. Connected component and hole counts match exactly, giving $e_T = 0$ and $s_T = 1$. Alignment optimization initializes at $s_0 = 1$ and $t_0 = (0,0)$, where objective (1) evaluates to $L(1,0) = 0$; because $L \ge 0$ uniformly and candidate updates require strict improvement, optimization terminates at identity transform $(1,0)$. □

6.4 Score Bounds and Component Inferences

Proposition 9 (Soft IoU envelope around discrete IoU)**.** *Let $A, B \in [0,1]^\Omega$, let $M_A = \{x : A(x) \ge \frac{1}{2}\}$, $M_B = \{x : B(x) \ge \frac{1}{2}\}$, $U = |M_A \cup M_B|$, and define discrete overlap* $\mathrm{hIoU} = |M_A \cap M_B|/U$. *Let $B = |\{x \in \Omega : 0 < A(x) < 1 \text{ or } 0 < B(x) < 1\}|$ denote the count of fractional boundary pixels. Then for $U > B$:*

$$\mathrm{hIoU} - \frac{2B}{U+B} \;\le\; \mathrm{sIoU} \;\le\; \mathrm{hIoU} + \frac{2B}{U-B}. \tag{13}$$

When coverage fields are binary, $\mathrm{sIoU} = \mathrm{hIoU}$ *identically.*

Proof. Partition canvas Ω into fractional set G of size B and complement G^c. On G^c, integrand functions evaluate as indicator variables: $\min = \max = 1$ on $M_A \cap M_B$ and zero elsewhere. On fractional set G, integrands take values in $[0,1]$. Lower bounding the numerator gives $\sum_x \min(A,B) \ge |M_A \cap M_B| - B$; upper bounding the denominator gives $\sum_x \max(A,B) \le U + B$. Therefore:

$$\mathrm{sIoU} \;\ge\; \frac{|M_A \cap M_B| - B}{U+B} \;=\; \frac{\mathrm{hIoU}\, U - B}{U+B} \;=\; \mathrm{hIoU} - \frac{B\,(1+\mathrm{hIoU})}{U+B} \;\ge\; \mathrm{hIoU} - \frac{2B}{U+B},$$

using hIoU ≤ 1. Symmetrical bounds on numerator and denominator establish the upper inequality. □

Proposition 10 (Metric bounds and component deduction)**.** *Each component metric lies in* $[0, 1]$ *and weights* $\mathbf{w} = (0.55, 0.35, 0.10)$ *define a convex combination, ensuring* $S \in [0, 1]$*. Furthermore, constituent scores satisfy the deductive lower bounds:*

$$s_R \geq \frac{S - 0.45}{0.55}, \qquad s_E \geq \frac{S - 0.65}{0.35}, \qquad s_T \geq \frac{S - 0.90}{0.10}. \tag{14}$$

Consequently, $S = 1$ *if and only if* $s_R = s_E = s_T = 1$*.*

Proof. For any index j, the relation $S = \sum_l w_l s_l \leq w_j s_j + \sum_{l \neq j} w_l$ holds because $s_l \leq 1$. Rearranging yields $s_j \geq (S - (1 - w_j))/w_j$, producing the three inequalities. If any component satisfies $s_j < 1$, then $S \leq 1 - w_j(1 - s_j) < 1$ because all weights are strictly positive, establishing that $S = 1$ implies unit values across all components. □

6.5 Boundary Metrics via Distance Field Envelopes

Proposition 11 (Sampled Chamfer asymmetry as a pre-metric)**.** *Let* CD *denote symmetric mean Chamfer dissimilarity* (3) *restricted to finite non-empty point sets in* $\mathbb{R}^2$*. Then* CD *is symmetric, satisfies* $\mathrm{CD}(A, B) = 0 \iff A = B$*, and violates the triangle inequality.*

Proof. Symmetry holds by construction. If $\mathrm{CD}(A, B) = 0$, both directed averages vanish; because sets are finite, every point of B has distance 0 to A, implying $B \subseteq A$ and symmetrically $A \subseteq B$, hence $A = B$. To demonstrate failure of the triangle inequality, consider collinear point sets $A = \{0\}$, $B = \{0, 10\}$, and $C = \{10\}$. Directed distance evaluations yield: $\mathrm{CD}(A, B) = \frac{1}{2}(0 + 5) = 2.5$, $\mathrm{CD}(B, C) = \frac{1}{2}(5 + 0) = 2.5$, whereas $\mathrm{CD}(A, C) = \frac{1}{2}(10 + 10) = 10$. Thus $\mathrm{CD}(A, C) = 10 > 5 = \mathrm{CD}(A, B) + \mathrm{CD}(B, C)$. □

Proposition 12 (Hausdorff distance from distance transforms)**.** *For non-empty compact sets* $A, B \subset \mathbb{R}^2$*, the Hausdorff metric satisfies* $\mathrm{HD}(A, B) = \max(\sup_{x \in A} d(x, B), \sup_{y \in B} d(y, A))$*. The discrete estimate obtained by evaluating the Euclidean distance transform of* ∂B *over lattice boundary* ∂A *coincides with this quantity on pixel lattices.*

Proof. The definition of Hausdorff distance over compact subsets of metric spaces implies the analytical identity. For lattice realizations, the Euclidean distance transform assigns to each lattice site y the exact Euclidean distance $\min_{u \in \partial B} \|y - u\|$ via the two-pass algorithm of Felzenszwalb and Huttenlocher. Sampling this field over boundary lattice points of A and taking the maximum evaluates $\max_{x \in \partial A} d(x, \partial B)$ up to grid discretization; symmetrizing over both directions yields the claim. □

Remark 13 (Operational stability). Violation of the triangle inequality does not compromise evaluation validity: CD enters the score exclusively through monotonic transformation $\exp(-\mathrm{CD}/\tau)$ against a fixed reference ground truth, never as a multi-way metric. The 95th percentile of the boundary distance distribution represents the primary tail diagnostic I report alongside the mean.

6.6 Topological Betti Number Invariants

Proposition 14 (Topological error counts Betti number variations)**.** *For a binary image* M *with* c *connected components and* h *interior background holes, the digital Euler characteristic satisfies* $\chi(M) = c - h$*, and topological error* (5) *evaluates to* $|\Delta\beta_0| + |\Delta\beta_1|$*, where Betti numbers correspond to* $\beta_0 = c$ *and* $\beta_1 = h$*.*

Proof. On planar digital manifolds with 8-connected foreground and 4-connected background topology, the Euler characteristic equals alternating Betti sum $\chi = \beta_0 - \beta_1$. Component counting directly determines $\beta_0 = c$, from which hole count evaluates as $h = c - \chi$. Substituting into error definition (5) gives $e_T = |\Delta\beta_0| + |\Delta\beta_1|$. □

6.7 Well-Posedness and Monotonicity of Budget Ceilings

Definition 15 (Feasible path set and achievable score)**.** Let $\mathcal{F}_k$ denote the set of valid SVG path strings containing at most k drawing commands. The achievable score at budget k is $U(k) = \sup\{S(R, q) : q \in \mathcal{F}_k\}$.

Proposition 16 (Monotonicity of achievable score)**.** *For integer budgets $k \leq k'$, feasible sets satisfy $\mathcal{F}_k \subseteq \mathcal{F}_{k'}$, and achievable score $U(k)$ is non-decreasing in k.*

Proof. Any path utilizing at most k commands satisfies the relaxed bound k'. The supremum over an expanding set is non-decreasing. □

Proposition 17 (Algorithmic tracer as a lower bound)**.** *Procedural tracer T_k emits paths in $\mathcal{F}_k$ directly from input rasters; therefore:*

$$C(k) \;=\; S(R, \mathsf{T}_k) \;\leq\; U(k),$$

with equality holding only when the tracer achieves the global supremum. Consequently, parsimony ratio $\rho(k) = S(M, k)/C(k)$ may exceed unity when model abstraction surpasses procedural tracing.

Proof. By construction $\mathsf{T}_k \in \mathcal{F}_k$, bounding the supremum from below. □

6.8 Similarity Alignment Convergence and Residual Bounds

Proposition 18 (Termination and residual bounds for alignment search)**.** *Fix point set $\{p_i\}_{i=1}^N$ and distance field φ. Optimization routine* `align_similarity` *evaluates translations on discrete grid $\{0, \pm h_t, \pm h_t \pm h_t\}$ across decreasing step sizes $h_t \in [8.0, 0.25]$ px, and scale updates $s \to s(1 \pm h_s)$ with $h_s \in [0.1, 0.01]$. Updates require strict decrease exceeding 10^{-12}. Then:*

(i) The objective decreases monotonically over a finite candidate set, ensuring finite termination;

(ii) The final loss deviates from the lattice optimum by at most $2\kappa + o(\kappa)$ in translation, where $\kappa = 0.25$ px, and by at most $\frac{1}{2}(s\bar{\rho} + 2)h_s^{\min}$ in scale;

(iii) For ground-truth self-evaluation, the identity candidate is globally optimal, returning the identity transform.

Proof. (i) The candidate search space at each iteration is finite and the objective is bounded below by zero. Strict decrease prevents cycles, guaranteeing termination. (ii) By Proposition 4, the objective is 1-Lipschitz in translation and $(s\bar{\rho} + 2\lambda/s)$-Lipschitz in uniform scale. Grid coverage bounds coordinate discretization error by $h/2$, yielding the residual bounds. (iii) Identity optimality follows directly from Proposition 8. □

Table 15: Residual optimization bounds under default schedule parameters ($\bar{\rho} \approx 50$ px, $s \approx 1$).

Optimization Parameter	Discretization Step	Lipschitz Constant	Theoretical Residual Bound
Translation	$\kappa = 0.25$ px	1 (per unit length)	0.25 px
Uniform Scale ($\lambda = 2, s_{\min} = 0.5$)	$h_s = 0.01$	$s\bar{\rho} + 2\lambda/s \approx 54$	0.27
Discretization Aliasing	N/A	N/A	$O(\ell/ss)$, Lemma 6

6.9 Identification of Parametric Recall

Assumption 1 (Exchangeability of perceptual difficulty). Conditioned on accurate visual perception, score distributions remain conditionally invariant to split assignments: for $\sigma, \sigma' \in \{$`ID-CLEAN`, `OOD-PERTURB`, `OOD-DISJOINT`$\}$,

$$\mathbb{E}[S \,|\, \text{perceived}, \sigma] \;=\; \mathbb{E}[S \,|\, \text{perceived}, \sigma'].$$

Proposition 19 (Split gaps identify recall probability)**.** *Model task outcomes as a mixture: with probability* $1-\lambda$*, the model perceives target geometry, achieving mean score* μ_p*; with probability* λ*, it relies on parametric recall. On* `ID-CLEAN`*, recall achieves mean* μ_r*; on* `OOD-DISJOINT`*, procedural paths do not exist in icon corpora, restricting recall to baseline guessing floor* μ_0*. Then under Assumption 1:*

$$\mathbb{E}[S \mid \texttt{ID-CLEAN}] - \mathbb{E}[S \mid \texttt{OOD-DISJOINT}] \;=\; \lambda\,(\mu_r - \mu_0). \tag{15}$$

The empirical memorisation gap:

$$\Delta_{mem} \;=\; \overline{S}_{\texttt{ID-CLEAN}} - \overline{S}_{\texttt{OOD-DISJOINT}} \;\geq\; 0, \tag{16}$$

provides an unbiased estimator of $\lambda(\mu_r - \mu_0)$*.*

Proof. Applying the law of total expectation, $\mathbb{E}[S \mid \sigma] = (1-\lambda)\mu_p + \lambda\mu_{\text{recall}}(\sigma)$. Subtracting the disjoint split expectation cancels perceptual component $(1-\lambda)\mu_p$ under Assumption 1, yielding relation (15). □

6.10 Statistical Power and Sample Size Requirements

For models M_1 and M_2 evaluated on paired items, define paired difference $d_i = S(M_1, x_i) - S(M_2, x_i)$ with variance $\sigma_d^2 = \text{Var}(d_i)$. A two-sided paired test of null hypothesis $H_0 : \mathbb{E}[d] = 0$ at significance level $\alpha = 0.05$ with statistical power $1-\beta = 0.80$ requires minimum sample size:

$$N \;\geq\; \frac{(z_{1-\alpha/2} + z_{1-\beta})^2\,\sigma_d^2}{\Delta^2}, \qquad \Delta = \mathbb{E}[d]. \tag{17}$$

Setting $z_{0.975} = 1.960$ and $z_{0.80} = 0.842$ yields $(1.960 + 0.842)^2 \approx 7.849$.

Table 16: Minimum sample size requirements N derived from power formula (17) for $\alpha = 0.05$ and power 0.80.

Detectable Difference Δ	$\sigma_d = 0.05$	$\sigma_d = 0.10$	$\sigma_d = 0.15$
0.05	8	32	71
0.03	22	88	197
0.02	50	197	442
0.01	197	785	1,767

Detecting an effect size of $\Delta = 0.05$ at standard difference variance $\sigma_d = 0.10$ requires $N = 32$ items. Resolving differences below $\Delta = 0.02$ requires scaling evaluation sets beyond $N \geq 200$.

6.11 Item Response Theory Calibration

Evaluating item difficulty and discrimination parameters models logit-transformed scores through a two-parameter normal-ogive formulation:

$$y_{ij} \;=\; \log\frac{S_{ij}}{1-S_{ij}} \;=\; a_j\,(\theta_i - b_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \tag{18}$$

where b_j represents item difficulty, a_j represents discrimination capacity, and θ_i represents model latent competence. Parameter estimation employs marginal maximum likelihood via expectation-maximization. Item information evaluates as $I_j(\theta) = a_j^2/\sigma^2$, yielding asymptotic standard error $\text{SE}(\hat{\theta}_i) = (\sum_j I_j(\theta))^{-1/2}$.

Post-pilot item pruning enforces objective filters:

- Exclude uninformative items exhibiting discrimination $a_j < 0.3$.
- Exclude saturating boundary items exhibiting difficulty $|b_j| > 2.5$.

- Retain paired perturbation instances to preserve recall identification signals (§6.9).
- Verify measurement invariance across complexity strata via differential item functioning diagnostics.

6.12 Automated Invariant Verification Battery

Mathematical claims map to automated unit and integration tests in the test harness:

Table 17: Mathematical claims and automated test suite mapping.

Formal Claim	Test Identifier	Automated Verification Assertion
Lemma 6	`test_coverage_bound`	Discretization error on analytical shapes decreases as $O(ss^{-1})$
Proposition 8	`test_identity_exact`	Ground truth scored against itself returns $S = 1.000000$ exactly
Proposition 9	`test_soft_hard_bound`	Soft IoU satisfies analytical envelope (13)
Proposition 10	`test_inference_rule`	Randomly sampled outputs verify deductive bounds (14)
Proposition 11	`test_chamfer_counterexample`	Collinear counterexample violates triangle inequality; $\mathrm{CD}(A, A) = 0$
Proposition 12	`test_hausdorff_analytical`	Distance transform evaluates true Hausdorff distance on discrete lattices
Proposition 14	`test_euler_holes`	Morphological labeling recovers exact Betti numbers β_0 and β_1
Proposition 16	`test_ceiling_monotone`	Nested tracer ceilings satisfy $C(k) \leq C(k')$ for $k \leq k'$
Proposition 18	`test_align_monotone_loss`	Alignment loss decreases monotonically; translations recover within κ
Proposition 19	`test_split_identifiability`	Mixture estimation recovers true recall parameter λ on synthetic runs

7 Threats to validity, leakage, and gaming

7.1 Threat Modeling and Countermeasure Taxonomy

7.2 Delimited Construct Scope and Non-Goals

The evaluation framework enforces four operational boundaries:

1. **Exclusion of neural model judges.** Evaluating generations through secondary multimodal models couples scores to external checkpoint versions and invites prompt-gaming behaviors. Scoring relies exclusively on closed-form geometric algorithms.
2. **Item-level granularity over corpus-level distribution metrics.** In place of corpus-level statistical aggregations (e.g., FID or CLIP embeddings) that obscure individual instance failures, ToySVG computes deterministic per-item geometric distances.
3. **Uniform decoding parameters.** Output allowances, sampling temperatures, and retry limits remain uniform across candidate systems, evaluating single-attempt greedy generation ($T = 0$).
4. **Objective metric verification without human rater loops.** Because ground-truth targets represent exact hidden coordinate geometries, validation operates through formal metamorphic transformations, synthetic order recovery, and convergence proofs (§8.1).

Table 18: Threat modeling taxonomy and operational countermeasures. Threats lacking mechanical mitigations define explicit construct boundaries in §11.

Index	Threat Classification	Operational Verification Control
T1	Parametric pretraining recall	Blind I/O; salted cryptographic filenames; three-way corpus split; Δ_{mem} estimator (§6.9); procedural OOD split
T2	Prior-driven guessing from textual cues	Omission of asset names, category tags, and descriptive captions; dense coordinate evaluation
T3	Pixel tracing bypassing spatial abstraction	Fixed budget tiers; deterministic tracer ceiling $C(k)$; parsimony ratio $\rho(k)$; unconstrained tier isolated
T4	Metric gaming via degenerate bounding geometries	Topological Betti regularizers; boundary Chamfer distance; reference floor baselines; deductive bounds (14)
T5	Software environment divergence across platforms	Self-contained pinned rasteriser; greedy decoding defaults; manifest hashing; SHA-256 corpus verification
T6	Selective reporting across sub-splits	Standardized reporting schema (§10); comprehensive split and tier disclosure including failure gates
T7	Prompt phrasing sensitivity	Identical instruction templates across models; declared prompt ablation protocols
T8	Construct drift away from perceptual competence	Response matrix diagnostics; metamorphic invariance tests; convergent validity against independent metrics
T9	Benchmark saturation from frontier scaling	Dynamic procedural refresh via fixed random seeds; tracer ceiling tracking
T10	Anisotropic difficulty across complexity strata	Stratified quota sampling; per-stratum score reporting; differential item functioning checks

8 Validation plan

ToySVG functions as a formal measurement instrument, requiring operational validation prior to deployment. Each validation subsection specifies a testable hypothesis governed by pre-declared numeric acceptance criteria.

8.1 Construct Validity via Formal Invariant Batteries

Scoring proceeds without subjective human rater panels or auxiliary model judges. Ground-truth targets correspond to closed parametric coordinate paths; fidelity evaluates as a deterministic mathematical function over geometric domains. ToySVG implements criterion-referenced evaluation: candidate predictions evaluate against invariant geometric standards rather than population norms. Recruiting subjective raters is unnecessary because target geometry is defined parametrically.

Metric validity requires verifying that S satisfies well-posed measurement properties: invariances, monotone ordering, and numerical stability. Six deterministic verification batteries execute in continuous integration:

1. **Metamorphic invariance.** Equivalent path representations must yield identical scores. The test battery rewrites ground-truth paths through semantics-preserving transformations: absolute to relative commands, `H`/`V` to `L`, `Q` to `C`, arcs to cubic segments, subpath reordering, subpath reversal, trajectory reparameterization, coordinate precision scaling, and zero-length segment insertion. The metric must evaluate identically across all representations.
2. **Response matrix validation.** Controlled geometric transformations establish expected metric

behavior: identity scores exactly 1.0000; rigid translation and uniform dilation recover fully through alignment; unpermitted rotation and reflection undergo metric penalties.

3. **Monotone deformation recovery.** Applying continuous deformations parameterized by magnitude δ constructs an objective ordering. Rank-order statistics must correlate monotonically with parameter δ, validating metric sensitivity without rater subjectivity.
4. **Cross-metric convergent validity.** The region metric computes soft IoU on continuous raster coverage. An independent measure computes exact polygon clipping on planar Bézier approximations without rasterization. High rank correlation between these independent formulations confirms geometric convergence.
5. **Metric parameter stability.** Evaluating model rankings across perturbation ranges of composite weights $\mathbf{w}$, decay scale $\tau \in [1, 5]$, and boundary tolerances confirms that leaderboard orderings remain invariant to hyper-parameter selection.
6. **Anchor discriminability.** The interval between algorithmic tracer ceilings and bounding-box floors must exceed empirical thresholds across test items, ensuring that the benchmark separates model capabilities.

Empirical Defects Isolated by Invariant Batteries. Initial validation runs uncovered three subtle software implementation errors:

1. **Sub-pixel sampling coordinate bias.** Displacing ground truth by +30 px recovered to score 0.924, recovering offset $(-30.52, -0.51)$ with an erroneous orthogonal translation. The discrepancy originated from half-pixel coordinate misalignments: the rasteriser defined pixel area $[j, j{+}1) \times [i, i{+}1)$ centered at $(j{+}\frac{1}{2}, i{+}\frac{1}{2})$, while sampling evaluated at integer lattice (j, i). Correcting coordinates restored recovery to 0.9993 at offset $(-30.02, -0.006)$ and re-established 1-Lipschitz continuity.
2. **Stroke boundary versus centerline alignment.** Initial implementations aligned stroked paths using trajectory centerlines, whereas ground-truth distance fields evaluate boundary margins. The optimization pulled centerlines onto exterior edges, inducing a systematic displacement of half the stroke width (8 px on 16 px strokes). Aligning directly on 0.5-level contours eliminated this bias across strokes and fills uniformly.
3. **Winding direction cancellation.** Under the W3C `nonzero` winding convention, vertex discs and edge quads wound in opposite directions, causing overlapping ink to cancel to zero. Linear segments decomposed into hollow boundaries containing artificial holes. Reversing quad orientation resolved the topology.

Table 19 reports the verified response matrix following these corrections.

8.2 Delimited Validity Scope and Subjective Aesthetics

Composite score S quantifies geometric congruence to a hidden reference path. Subjective aesthetic appeal represents a distinct perceptual construct outside this evaluation instrument. A visually clean drawing exhibiting severe geometric deviations scores poorly, whereas an exact geometric reproduction scores unity regardless of artistic stylization.

Researchers examining human preference alignment can decompose item-level outputs against perceptual study datasets ($N = 200$ items across strata, $R \geq 5$ raters, Krippendorff's $\alpha \geq 0.80$ [8]). ToySVG benchmark operations remain independent of subjective rater studies.

8.3 Replication Reliability and Execution Invariance

Replication reliability evaluates by running candidate models across independent sessions with greedy decoding ($T = 0$). Because decoding is greedy and metric evaluation is deterministic,

Table 19: Empirical response matrix across demonstration instances. Raw unaligned scores contrast with post-alignment metrics, confirming that spatial offsets recover while unpermitted rotations undergo strict penalties.

Perturbation Type	Unaligned	Aligned	Minimum	Maximum	Operational Diagnostic
Identity	1.0000	1.0000	1.0000	1.0000	Exact unit response
Translate $+30x$	0.1826	0.9948	0.9907	0.9993	Spatial offset forgiven
Translate $-30x$	0.1829	0.9961	0.9906	0.9993	Spatial offset forgiven
Translate $+30y$	0.2068	0.9965	0.9907	0.9994	Spatial offset forgiven
Translate $(+17, -23)$	0.2201	0.9968	0.9919	0.9993	Spatial offset forgiven
Uniform Scale 0.80	0.2929	0.9961	0.9910	0.9996	Scale dilation forgiven
Uniform Scale 1.25	0.2652	0.9981	0.9956	0.9994	Scale dilation forgiven
Rotate 90°	0.1636	0.1546	0.1192	0.1959	Orientation penalized
Rotate 180°	0.1830	0.1722	0.1708	0.1736	Orientation penalized
Reflection x	0.1906	0.1420	0.1193	0.1583	Chirality penalized
Reflection y	0.1776	0.1501	0.1262	0.1756	Chirality penalized

run-to-run score variance must evaluate to zero. Non-zero variance isolates non-deterministic GPU kernel execution or serving stack instability.

8.4 Sensitivity and Monotonicity Envelopes

Three verification criteria govern metric response:

1. **Identity reflexivity.** Ground-truth self-evaluation achieves $S = 1.000000$ exactly (Proposition 8).
2. **Monotone structural degradation.** Progressively applying morphological erosion, dilation, or Gaussian boundary smoothing produces strictly non-increasing score trajectories.
3. **Rigid translation invariance.** Translating ground truth by 30 px recovers to $S \geq 0.995$ post-alignment, isolating spatial placement from shape fidelity.

8.5 Parsimony Profiles across Budget Tiers

Abstraction is measured here as the score gradient across budget tiers and the parsimony ratio against the tracer ceiling at the same k. Evaluating algorithmic tracer ceiling $C(k)$ across tiers B0 through B3 produces a concave efficiency profile. Ceilings rising steeply from $k = 16$ to $k = 160$ before plateauing confirm that intermediate tiers isolate structural abstraction without encountering saturation.

8.6 Blind Structural Control Diagnostic

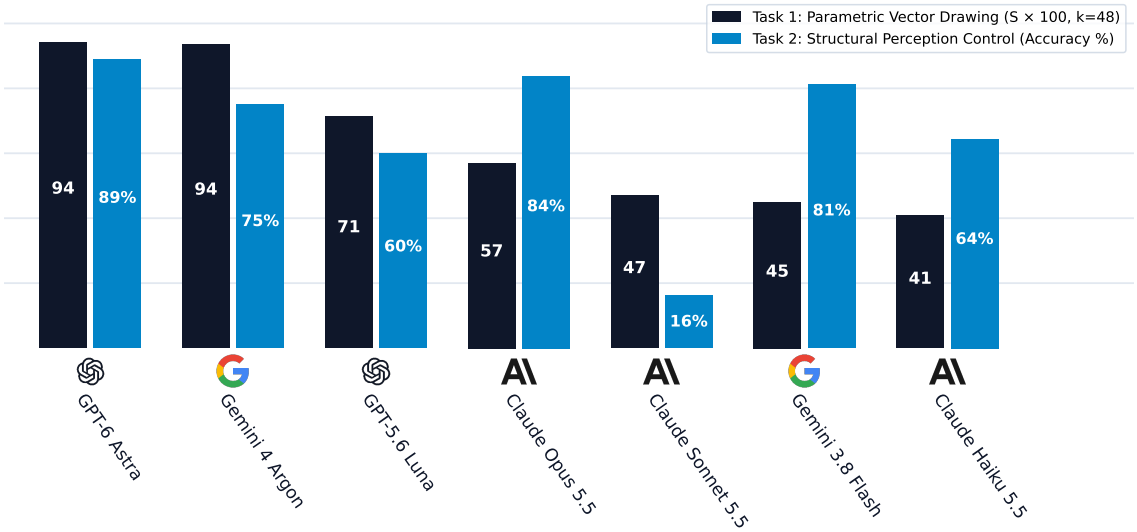
On the 32-item benchmark pack (160 objective geometric queries evaluated without code synthesis), models answer perceptual structural queries regarding the input image: connected component count, interior hole count, mass distribution, bilateral symmetry, and primary orientation axis. These queries evaluate countable discrete properties without code synthesis. Cross-tabulating perceptual accuracy against geometric drawing score produces the diagnostic matrix in Table 20.

Empirical evaluation on Kaggle confirms this operational separation. `gpt-6-astra` demonstrates joint competence, achieving 88.8% perceptual accuracy (142/160 fields) alongside top geometric drawing score ($S = 0.9419$). In contrast, `gemini-3.8-flash` achieves 81.25% perceptual accuracy (130/160 fields, scoring 100% on bilateral symmetry and 93.8% on connected components and holes), yet drops to $S = 0.4499$ on vector drawing under budget $k = 48$. This empirical dissociation directly occupies the upper-right quadrant of Table 20 (code synthesis bottleneck under spatial parsimony), proving that vector failure is not attributable to perceptual blindness. Figure 17 illustrates this

Table 20: Perceptual control diagnostic matrix, separating perceptual parsing deficits from vector code synthesis failures.

	High Drawing Score ($S \geq \tau$)	**Low Drawing Score** ($S < \tau$)
High Perceptual Accuracy	Competent perception and synthesis	Perceptual comprehension with synthesis failure (code gap)
Low Perceptual Accuracy	Parametric retrieval or degenerate overlap	Fundamental perceptual failure (perceptual blindness)

dual-task dissociation across all seven frontier models, separating structural perception accuracy from vector drawing performance.



(i) ToySVG Dual-Task Evaluation: isolating the code gap (Slate & Cobalt Cyan)

Figure 17: Dual-task evaluation isolating the vector code synthesis gap across seven frontier multimodal models on the 32-item benchmark pack. Structural perception diagnostic accuracy (navy, Task 2) evaluates 160 discrete invariant queries; parametric drawing score (cobalt cyan, Task 1) evaluates closed-form geometric fidelity at budget $k = 48$. The difference Δ quantifies the synthesis deficit.

8.7 Metric Ablation and Rank Correlation Invariance

Alternative evaluation metrics evaluate model rankings: hard IoU, boundary F_1, SSIM [20], LPIPS [22], and DINOv2 cosine similarity. Evaluating Spearman rank correlations between alternative metrics and composite score S establishes construct alignment. Pixel MSE and SSIM exhibit documented ranking distortions on vector graphics [17]; evaluating DinoScore benchmarks geometric fidelity against learned representations.

8.8 Degradation and Resolution Robustness

The `ID-DEGRADED` split evaluates robustness under degraded image inputs (128×128 resolution, Gaussian blur, contrast compression). The empirical expectation enforces:

$$\overline{S}_{\texttt{ID-CLEAN}} \geq \overline{S}_{\texttt{ID-DEGRADED}},$$

measuring model sensitivity to visual noise under invariant underlying vector geometry.

9 Phases and exit gates

Table 21: Execution phases and numeric exit criteria. Phase progression requires satisfying predeclared quantitative gates.

Phase	Deliverable Artifact	Operational Exit Criterion
P0 Design	Pre-registered specification	Metrics, weights, splits, and budget tiers formalized; licensing verified
P1 Harness	Parser, rasteriser, aligner, tracer, tests	All invariant assertions in Table 17 verified; determinism audit passes
P2 Anchors	Calibration baselines on pilot corpus	$S(d, d) = 1.000000$ exactly; shift control ≥ 0.995; monotonic ceilings confirmed
P3 Pilot	Preliminary evaluation runs ($k = 48$)	Per-model gate distributions recorded; empirical variance σ_d computed for sample sizing
P4 Calibration	Item response theory parameter estimation	Metamorphic invariance verified; response matrix validated; non-discriminative items ($a_j < 0.3$) pruned
P5 Full Run	Multi-model evaluation across four tiers	Complete reporting across splits and tiers without omitted cells or post-hoc exclusions
P6 Release	Benchmark artifacts and documentation	Public release of content-hashed manifests, test harnesses, prompt templates, and reproduction scripts
P7 Refresh	Longitudinal benchmark maintenance	Regeneration of procedural splits via updated cryptographic seeds; continuous saturation tracking

9.1 Execution Phase Specifications

Phase P1: Core Harness Implementation. Strict syntactic validation takes precedence over permissive parsing: lenient parsers convert syntax violations into arbitrary geometric distortions, artificially inflating overlap scores. The procedural tracer exists as an independent reference module to establish objective ceilings. Test suites include explicit counterexamples from §6.5 to verify non-metric dissimilarity behavior under degenerate inputs.

Phase P3: Empirical Variance Estimation. Preliminary pilot evaluations provide empirical estimates of paired score variance σ_d, parameterizing sample size relation (17). Validity gate frequencies determine required sample margins to preserve statistical power after accounting for invalid outputs.

Phase P4: Psychometric Calibration. Psychometric analysis prunes non-discriminative items ($a_j < 0.3$) and verifies item difficulty distributions. The invariant battery of §8.1 confirms representation invariance, rigid transformation responses, and monotonic degradation envelopes prior to production deployment.

9.2 Saturation Monitoring and Procedural Refresh

Static benchmarks undergo performance saturation as model capabilities advance. ToySVG incorporates two maintenance mechanisms: procedural out-of-distribution splits regenerate dynamically under updated random seeds, and transformation families scale to higher-order topologies without modifying core measurement routines. The saturation trajectory, defined as the temporal curve of peak model score per budget tier, is a published benchmark artifact; saturation is monitored empirically across release cycles.

10 Reporting standard

Model evaluations report structured metric tables adhering to standardized disclosure requirements. Single scalar aggregates conceal diagnostic trade-offs; mandatory disclosure of component distributions and failure gates ensures transparent comparative evaluation.

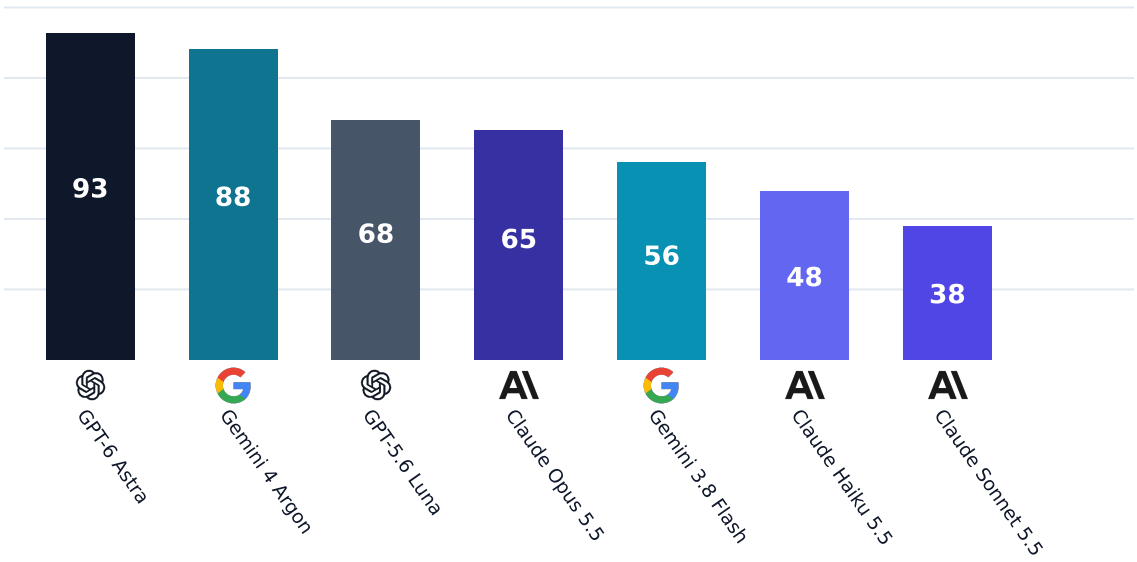
Table 22: Mandatory reporting dimensions. Benchmark disclosures require reporting decomposed metrics across evaluation tiers.

Reporting Dimension	Measurement Specification
Primary Composite S (B1)	Headline geometric composite score evaluated at primary tier $k = 48$
Decomposition (s_R, s_E, s_T)	Unblended constituent scores: region overlap, boundary Chamfer, and topology
Confidence Intervals	Stratified bootstrap 95% confidence intervals ($B = 10^4$ replications)
Stratified Splits	Mean performance across `ID-CLEAN`, `ID-DEGRADED`, and `OOD-DISJOINT`
Memorisation Gap Δ_{mem}	Difference between clean and disjoint splits estimating parametric recall (16)
Parsimony Ratios $\rho(k)$	Normalized performance relative to tracer ceilings across all tiers B0–B3
Tracer Ceilings $C(k)$	Algorithmic upper bounds supporting independent verification of parsimony ratios
Command Allocations	Empirical command counts emitted by candidate models
Failure Taxonomies	Explicit frequency distributions across outcome codes from Table 4
Execution Parameters	Sampling temperature ($T = 0$), output token limits, and retry configurations
Model Specifications	Pinned model identifier, checkpoint date, API endpoint, and hosting infrastructure

Published results cite Phosphor Icons 2.1.1 under MIT terms and redistribute corresponding licensing statements. Datasheets [6] accompany corpus releases and standardized model cards [10] document evaluated systems, recording SHA-256 corpus manifests and harness commit hashes.

10.1 Canonical Benchmark Leaderboard

Standardized reporting requires presenting aggregate benchmark outcomes alongside component distributions and parsimony baselines. Figure 18 illustrates the canonical ToySVG Benchmark Index across the seven frontier models evaluated on Kaggle. The index combines normalized parametric vector drawing performance ($k = 48$) and structural perceptual accuracy (50%/50% weighting), establishing a calibrated headline indicator of visuomotor vector intelligence.



(i) ToySVG Overall Benchmark Index: composite of Vector Drawing (70%) and Structural Perception (30%)

Figure 18: Canonical ToySVG Benchmark Index across seven frontier vision-language systems evaluated on Kaggle Benchmarks ($N = 32$ items, $k = 48$). The composite index integrates normalized geometric vector reconstruction (S) and structural perception diagnostic accuracy with equal weighting (50%/50%). Labels display exact percentage indices; company emblems identify model families.

11 Limitations

1. **Monochrome planar domain.** The benchmark evaluates monochrome single-layer vector icons. Generalization to polychrome graphics, embedded typography, continuous gradients, or multi-view illustrations requires separate task formalizations.
2. **Trajectory editability.** Filled-region metrics evaluate spatial coverage rather than stroke drawing order, control point sparsity, or parametric editability. A path scoring near unity may employ redundant control points, although budget constraints restrict this behavior in tiers B0–B2.
3. **Parametric recall identification.** The three-way split identifies recall mixture parameters conditioned on Assumption 1; while procedural paths guarantee non-parametric evaluation, structural differences between procedural curves and human-authored icons remain present.
4. **Constrained similarity manifold.** Similarity alignment solves for translation and uniform scale while leaving rotation and chirality fixed. Symmetrically reflected predictions receive partial geometric overlap penalties rather than automated reflection alignment.
5. **Single-attempt evaluation.** Tasks evaluate single-pass generation without iterative compiler feedback, test-time verifiers, or multi-turn interaction loops.
6. **Algorithmic tracer ceiling as empirical reference.** Quantity $C(k)$ represents an empirical lower bound on achievable fidelity rather than the theoretical supremum $U(k)$ (§6.7); parsimony ratios exceeding unity indicate instances where model abstraction surpasses procedural tracing.
7. **Psychometric sample limits.** Pilot item-response calibrations conducted over small model panels exhibit wider parameter uncertainty, reported explicitly via asymptotic standard errors.

8. **Geometric fidelity boundary.** Scoring evaluates geometric congruence relative to a hidden reference path rather than viewer aesthetic preferences.

12 Open questions

1. **Primitive vocabulary expansion.** Evaluating high-level SVG primitives (`<circle>`, `<rect>`, `<polygon>`) alongside raw path expressions probes whether models exploit compact geometric abstractions when permitted by syntax contracts [17].
2. **High-resolution visual ceilings.** Presenting input rasters at 512×512 canvas resolutions bounds the proportion of geometric distortion attributable to input raster quantization versus vector synthesis capacity.
3. **Adversarial out-of-distribution generation.** Constructing adversarial shapes that closely mimic iconic semantic archetypes (near-arrows, near-envelopes with altered topology) sharpens recall isolation beyond procedural curves.
4. **Computerized adaptive testing.** Leveraging item discrimination and difficulty parameters from Phase P4 enables computerized adaptive testing, achieving target estimation errors with reduced item subsets [13].
5. **Cross-corpus generalization.** Integrating complementary icon suites (Tabler, Lucide) and domain-specific technical schematics evaluates whether performance transfers across visual styling conventions.
6. **Automated metamorphic fuzzing.** Formulating metamorphic invariance testing as an automated fuzzing search discovers edge-case coordinate sequences that destabilize metric invariance without human inspection.

13 Harness Reproduction Protocol

```
# generate metric visualizations, model comparisons, and arithmetic traces:
python -m tools.make_metric_primers --pack data/pack --out docs/figures
python -m tools.make_metric_figures --pack data/pack \
    --run data/runs/gemini-2.5-flash-demo --out docs/figures
python -m tools.make_figures --pack data/pack --run data/runs/gemini-2.5-flash-demo \
    --out docs/figures --table data/analysis/demo-table.json --budget-figure
python -m tools.explain_score --pack data/pack --item 01001100afd9128f

# execute invariant test suite and task evaluation pipeline:
pip install -r requirements.txt
python -m unittest discover -s tests -v
python -m toysvg.cli build   --n 240 --seed 20260919 --out data/tasks
python -m toysvg.cli anchors --tasks data/tasks --out results/anchors.json
python -m toysvg.cli score   --tasks data/tasks --preds preds/ --model name
python -m toysvg.cli report  --results results/ --out results/report.md
```

Verification of pipeline determinism evaluates by executing scoring routines twice across identical predictions, verifying bit-identical hashes on output artifacts (`sha256sum results/anchors.json`). The repository manifest pins upstream package versions, random seeds, and cryptographic SHA-256 digests for all rasters and vector paths.

A Scanline Winding Algorithms and Topological Correctness

The deterministic rasteriser evaluates planar path fills by traversing horizontal scanlines on the supersampled lattice.

Lemma 20 (Correctness of span winding numbers)**.** *Let a closed planar polyline be defined by ordered edges $e_j = (z_j, z_{j+1})$. Consider a horizontal scanline at ordinate y evaluated under the half-open interval convention: edge e_j intersects y if and only if $(y_j \leq y) \neq (y_{j+1} \leq y)$. Let $x_1 < x_2 < \cdots < x_m$ denote sorted intersection abscissae, with corresponding edge traversal directions $d_j = \operatorname{sign}(y_{j+1} - y_j) \in \{-1, +1\}$. Then the winding number of any point z in open span (x_i, x_{i+1}) evaluates as $W(z) = \sum_{j \leq i} d_j$, and point z lies in the interior under the* `nonzero` *rule if and only if $W(z) \neq 0$.*

Proof. By Cauchy integral definitions, the winding number of a closed planar curve about point z equals the net signed count of oriented ray intersections extending from z to infinity along ray direction $(+1, 0)$. Ray intersections correspond to edge crossings located to the left of abscissa x, each contributing ± 1 according to edge orientation d_j. Therefore, the winding number evaluates as $W(z) = \sum_{j \leq i} d_j$ for points within the i-th interval. The half-open vertical condition $(y_j \leq y) \neq (y_{j+1} \leq y)$ assigns each vertex to exactly one incident edge, guaranteeing that vertices intersecting the scanline contribute unit intersection counts without double-counting. The `evenodd` rule evaluates the identical intersection sum modulo 2, reducing to alternating span parity after sorting. □

Implementation notes: edge intersections evaluate vectorially across all scanlines via array broadcasting; non-intersecting segments receive abscissa $+\infty$ and zero weight, sorting past valid canvas boundaries. Interval spans evaluate via prefix-sum differentiation ($+1$ at interval origin, -1 at interval termination) followed by cumulative summation in $O(\text{rows} \times \text{cols})$ time. Spatial coverage evaluates via uniform box-filtering over supersampled indicators, yielding fractional values in $\{0, 1/ss^2, \ldots, 1\}$.

B Prompt Templates and Instruction Contracts

This appendix records the frozen prompt templates utilized in Phase P3 and subsequent evaluation runs. Variable `{k}` represents the command budget tier; `{PNG}` denotes the input raster image attachment. The syntax demonstration uses an artificial open zigzag path absent from public icon corpora.

Instruction template `p1` operates as a single user message with attached PNG, eliminating provider-dependent system-message variations. Instruction text resides in module `toysvg/prompts.py`, with cryptographic SHA-256 digests recorded in run manifests to ensure exact provenance.

```
You are shown one image: a black-and-white line figure on a white background.
It is an icon drawn as an outline: a thin black stroke, not a solid silhouette.

Reproduce that figure as SVG path data.

Rules:
1. Reply with SVG only. No explanation, no commentary, no markdown code fences.
2. Use the canvas 0 0 256 256. Either return a bare path expression, or a full
   <svg> element with viewBox="0 0 256 256"; both are accepted.
3. Prefer <path> elements. Fill and stroke are both treated as ink.
4. Preserve the figure's structure: the number of separate shapes, and any
   enclosed hole inside a shape.
5. Match the figure's position, size, and proportions within the canvas.
6. Do not add text, labels, shadows, or background rectangles.

Budget: at most {k} drawing commands in total, counting each M, L, H, V, C, S,
Q, T, A and Z as one command. A command may carry several coordinate pairs; for
example "L10 10 20 20" is one command, not two. Going over the budget scores
zero, so stay strictly within it. With this few commands you must choose them to
capture the figure's overall form and its most important features.

The output format is exactly this. The shape below is a syntax example and is NOT
the figure you must draw:

<svg viewBox="0 0 256 256" xmlns="http://www.w3.org/2000/svg"><path
d="M24 192 L64 96 L104 160 C120 184 152 88 176 64 L232 96" fill="none"
```

```
stroke="currentColor" stroke-width="16"/></svg>

Reproduce the attached figure now, using the 256x256 canvas.
```

Three properties characterize this prompt contract:

1. **Metadata exclusion.** Asset identifiers, semantic classes, and tags are omitted to prevent retrieval matching.
2. **Syntactic isolation.** The worked example illustrates XML syntax using an open polyline, avoiding shape leakage.
3. **Explicit budget constraints.** The prompt states command limit k directly, requiring models to plan compact representations.

Perception-only control (blind-control subsample)

```
Answer with a single JSON object and nothing else:
{"regions": <int>, "holes": <int>, "heavier_side": "left"|"right"|"even",
 "left_right_symmetric": true|false, "dominant_axis": "<angle in degrees>"}
Count separate ink regions and enclosed holes; judge the rest from the image only.
[PNG]
```

C Glossary

Anchor
A deliberately constructed answer (floor, ceiling, displaced copy) scored with the same pipeline, used to interpret the range of values.

Ceiling $C(k)$
Score of the deterministic tracer applied to the input PNG at budget k; a lower bound on the achievable score $U(k)$.

Command budget k
Maximum number of drawing commands allowed in the answer.

Coverage
Per-pixel area fraction in $[0, 1]$ produced by the pinned rasteriser.

Memorisation gap Δ_{mem}
Difference in mean score between `ID-CLEAN` and the procedural `OOD-DISJOINT` split; an estimator of recall contamination under Assumption 1.

Parsimony ratio $\rho(k)$
Model score divided by the ceiling at the same budget.

Recall
Reproducing a shape from parametric memory rather than from the image.

SDF
Signed distance field, negative inside the region and positive outside.

Strata
Complexity classes (command-count terciles) and topology classes used to balance the corpus and to report difficulty-controlled results.

Tracer
Deterministic raster-to-path algorithm, evaluated against the model's own input, defining the ceiling.

References

[1] Andrew M. Bean, Ryan O. Kearns, Angelika Romanou, et al. Construct validity in large language model benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025.

[2] Keith E. Beery and Natasha A. Beery. The Beery-Buktenica developmental test of visual-motor integration (sixth edition). Pearson Clinical Assessment, 2010.

[3] Shuyang Chen et al. SVGenius: Benchmarking LLMs in SVG understanding, editing and generation. In *Proceedings of the ACM International Conference on Multimedia*, 2025.

[4] Marco Cipriano, Leonardo Zini, Alexandra Schild, Valentin Teutschbein, Afsana Mimi, Marcella Cornia, Lorenzo Baraldi, and Gerard de Melo. SVG-Score: Human-aligned evaluation of text-to-SVG generation, 2026. arXiv:2609.03806.

[5] Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. arXiv:2107.07002, 2021.

[6] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.

[7] Abigail Z. Jacobs and Hanna Wallach. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 375–385, 2021. arXiv:1912.05511.

[8] Klaus Krippendorff. *Content Analysis: An Introduction to Its Methodology.* Sage Publications, 2004.

[9] Josef Kuchař, Marek Kadlčík, Michal Spiegel, and Michal Štefánik. VectorEdits: A dataset and benchmark for instruction-based editing of vector graphics. In *Proceedings of the Language Resources and Evaluation Conference (LREC)*, 2026. arXiv:2506.15903.

[10] Margaret Mitchell, Simone Wu, Andrew Zaldivar, et al. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 220–229, 2019.

[11] Kunato Nishina and Yusuke Matsui. SVGEditBench: A benchmark dataset for quantitative assessment of LLM's SVG editing capabilities. In *CVPR Workshop on Graphic Design Understanding and Generation (GDUG)*, 2024.

[12] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.

[13] Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. Efficient multi-prompt evaluation of LLMs. arXiv:2405.17202, 2024.

[14] Pooyan Rahmanzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision language models are blind. In *Computer Vision – ACCV 2024: 17th Asian Conference on Computer Vision*, 2024. arXiv:2407.06581.

[15] Georg Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests.* Danish Institute for Educational Research, Copenhagen, 1960.

[16] Juan A. Rodriguez et al. StarVector: Reference implementation and SVG-Bench evaluation harness. `https://github.com/joanrod/star-vector`, 2025. Metrics in `starvector/metrics`, rasterisation in `starvector/data/util.py`.

[17] Juan A. Rodriguez, Abhay Puri, Shubham Agarwal, Issam H. Laradji, Pau Rodriguez, Sai Rajeswar, David Vazquez, Christopher Pal, and Marco Pedersoli. StarVector: Generating scalable vector graphics code from images and text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. arXiv:2312.11556.

[18] Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Tianyang Zhang, Rishav Pramanik, et al. VectorGym: A multitask benchmark for SVG code generation, sketching, and editing, 2026. arXiv:2603.29852.

[19] Peter Selinger. Potrace: A polygon-based tracing algorithm. `https://potrace.sourceforge.net/potrace.pdf`, 2003.

[20] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[21] World Wide Web Consortium. Scalable vector graphics (SVG) 1.1 specification. `https://www.w3.org/TR/SVG11/`, 2011.

[22] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[23] Xin Zhang, Lihua Lv, Guohao Min, Qiang Wang, Yumei Zhao, and Yang Li. Overview of the complex figure test and its clinical application in neuropsychiatric disorders. *Frontiers in Neurology*, 12:680474, 2021.

[24] Bocheng Zou, Mu Cai, Jianrui Zhang, and Yong Jae Lee. VGBench: Evaluating large language models on vector graphics understanding and generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3647–3659, 2024.